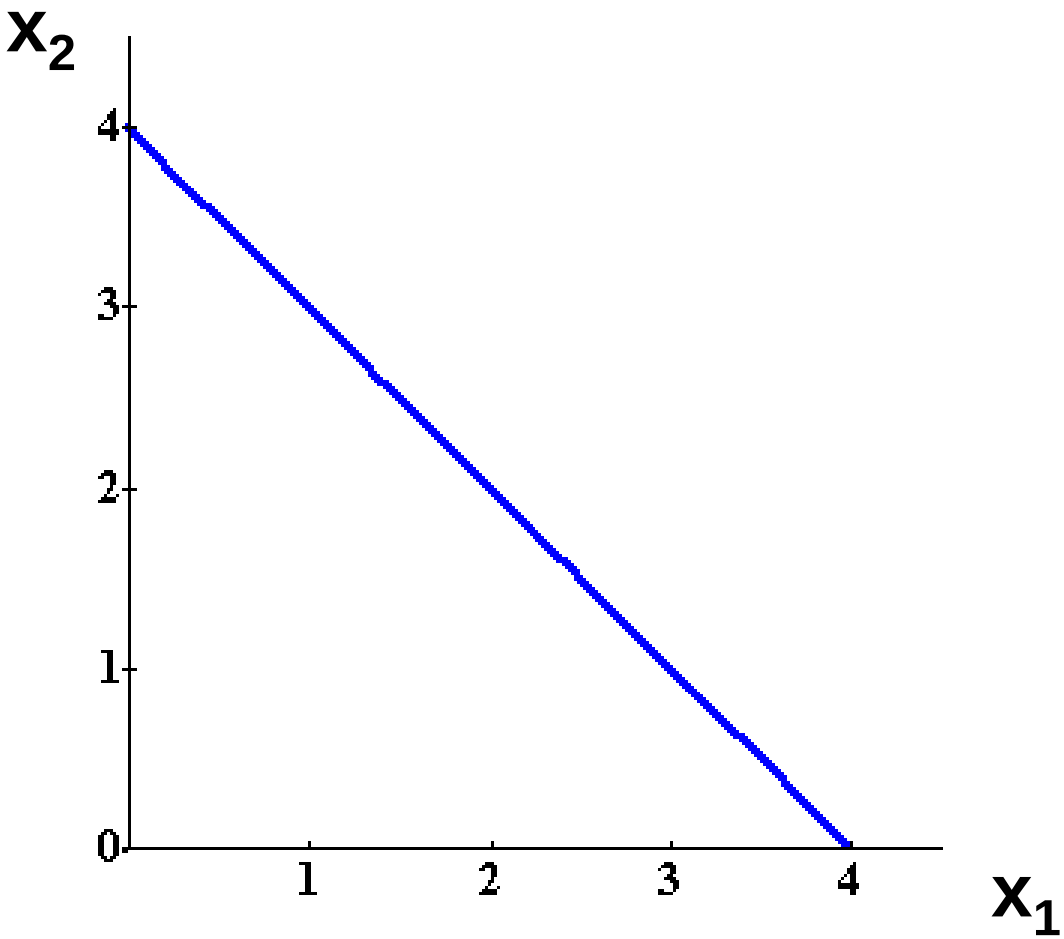
# Chapter Five

## Choice

# Economic Rationality

- **Behavioral postulate: The decision maker chooses its most preferred alternative from those available to it.**
- **The choice set consists of the available choices.**

# Rational Constrained Choice: An Example



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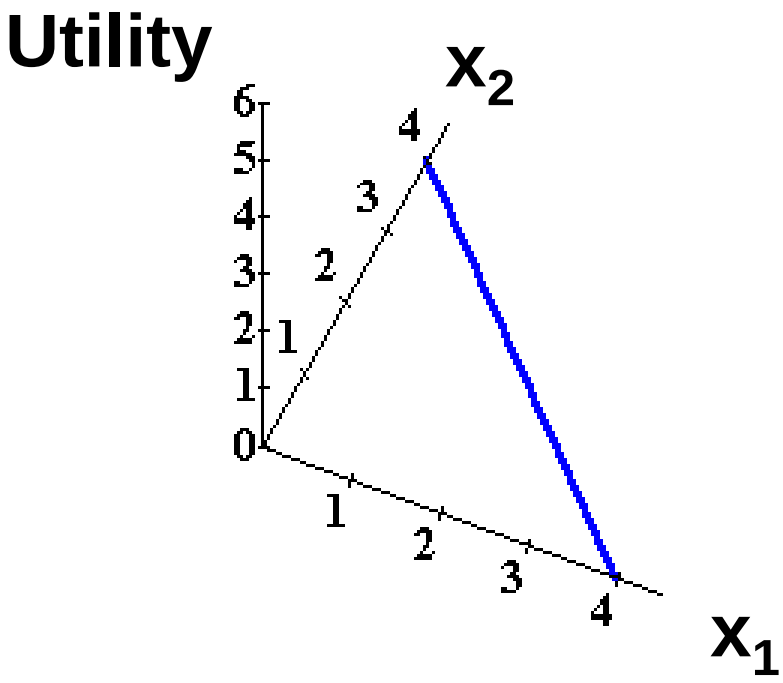

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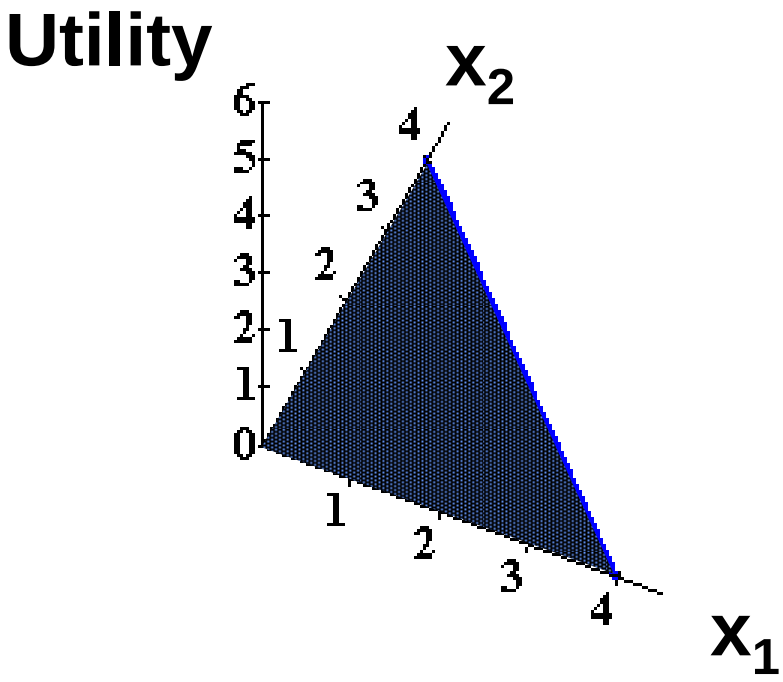

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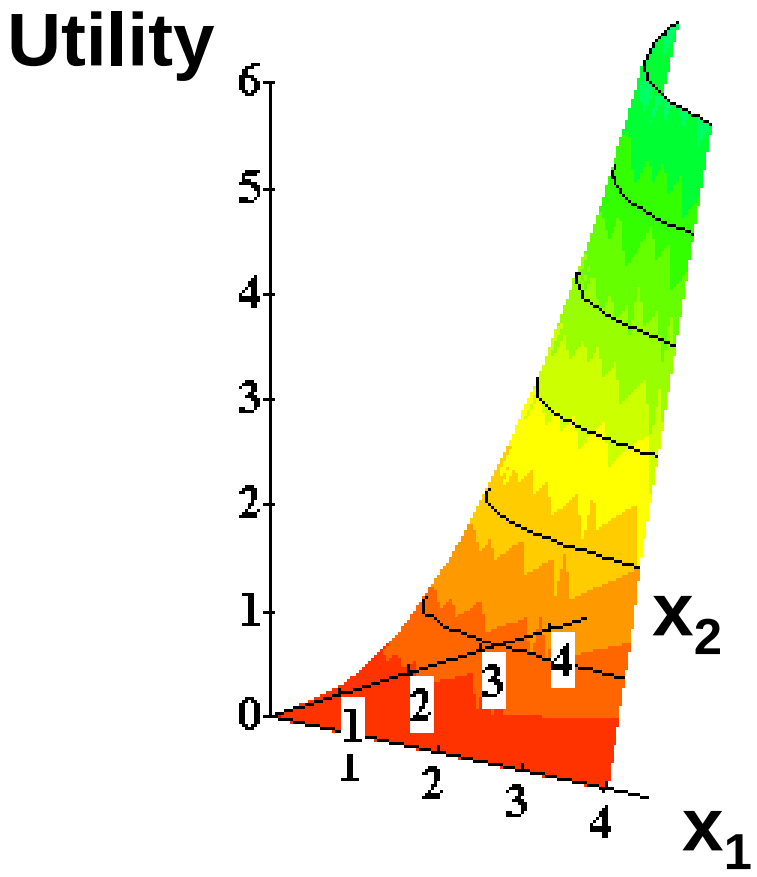

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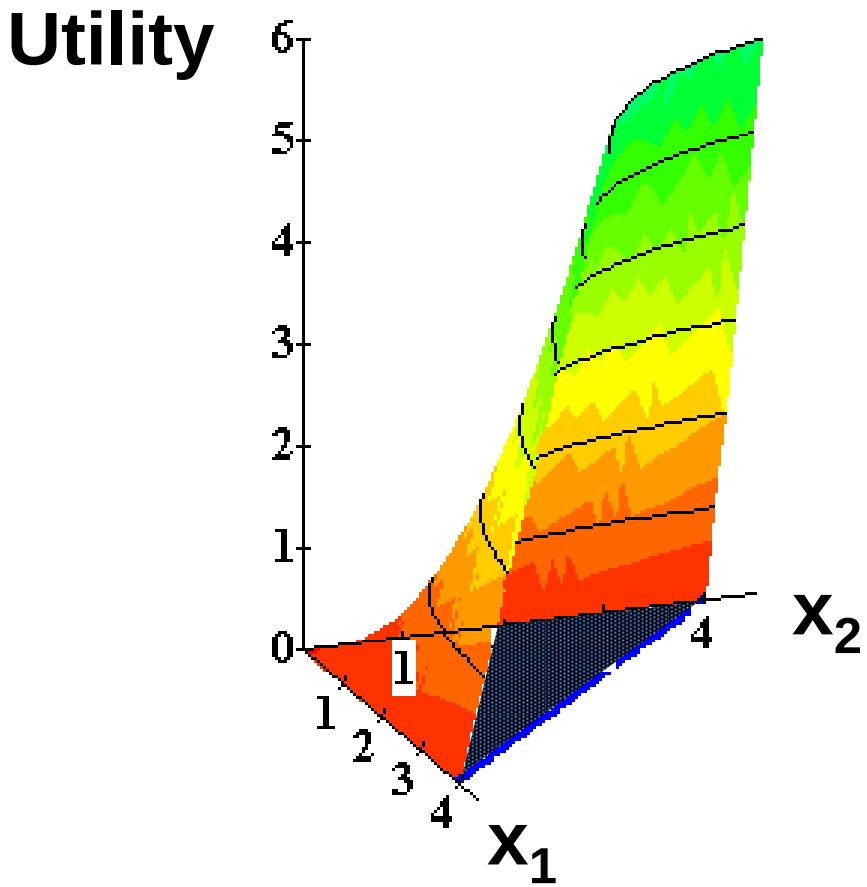

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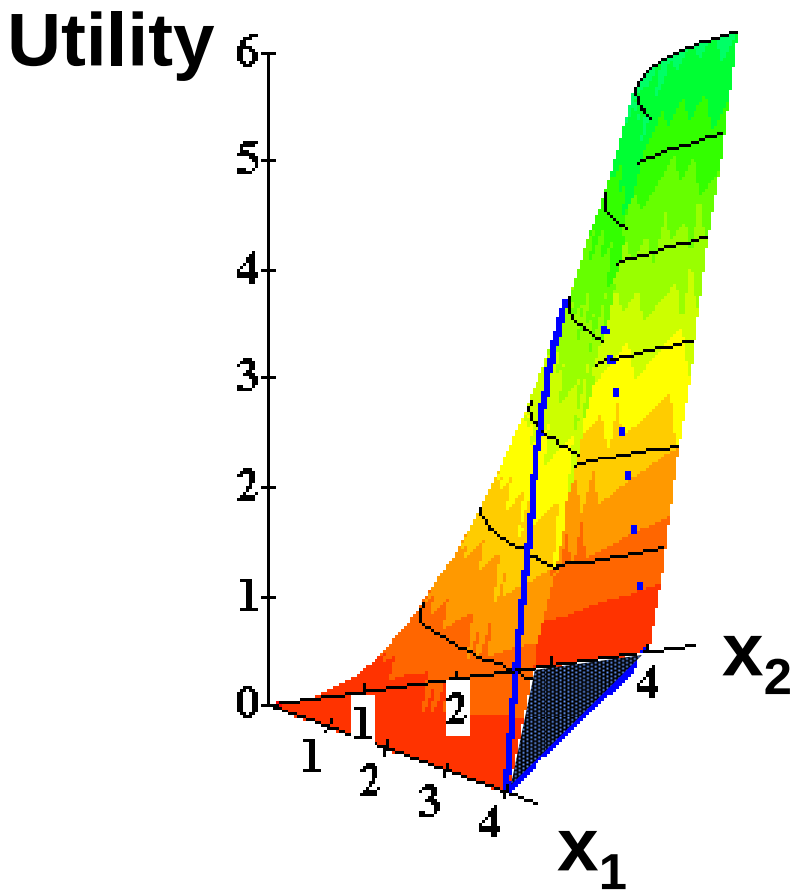

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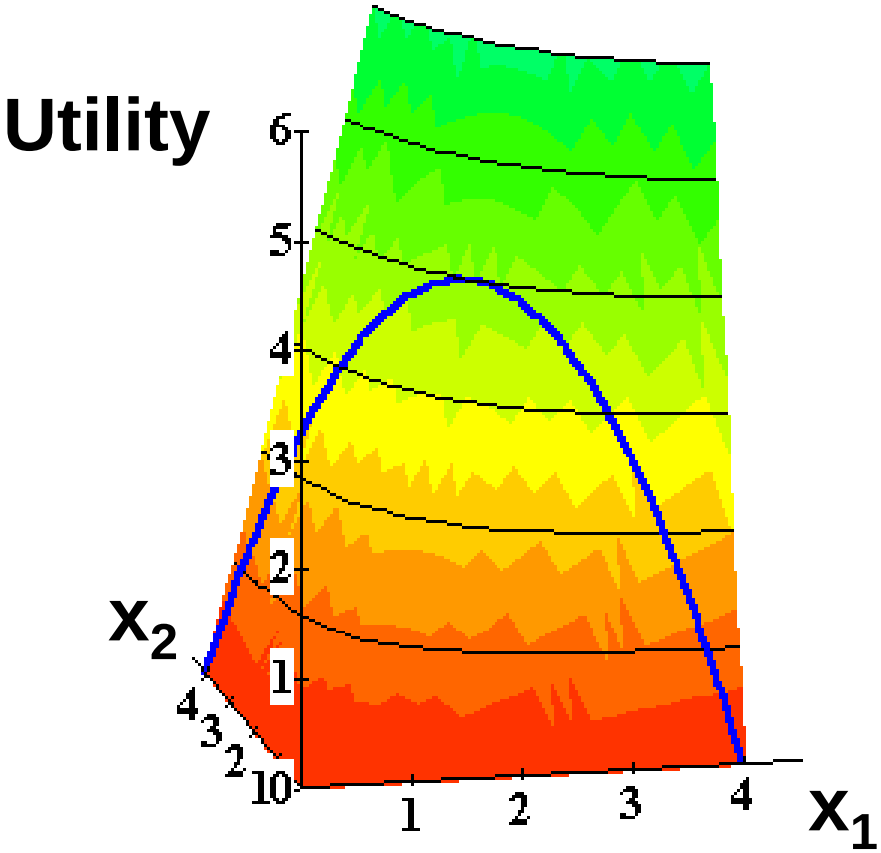

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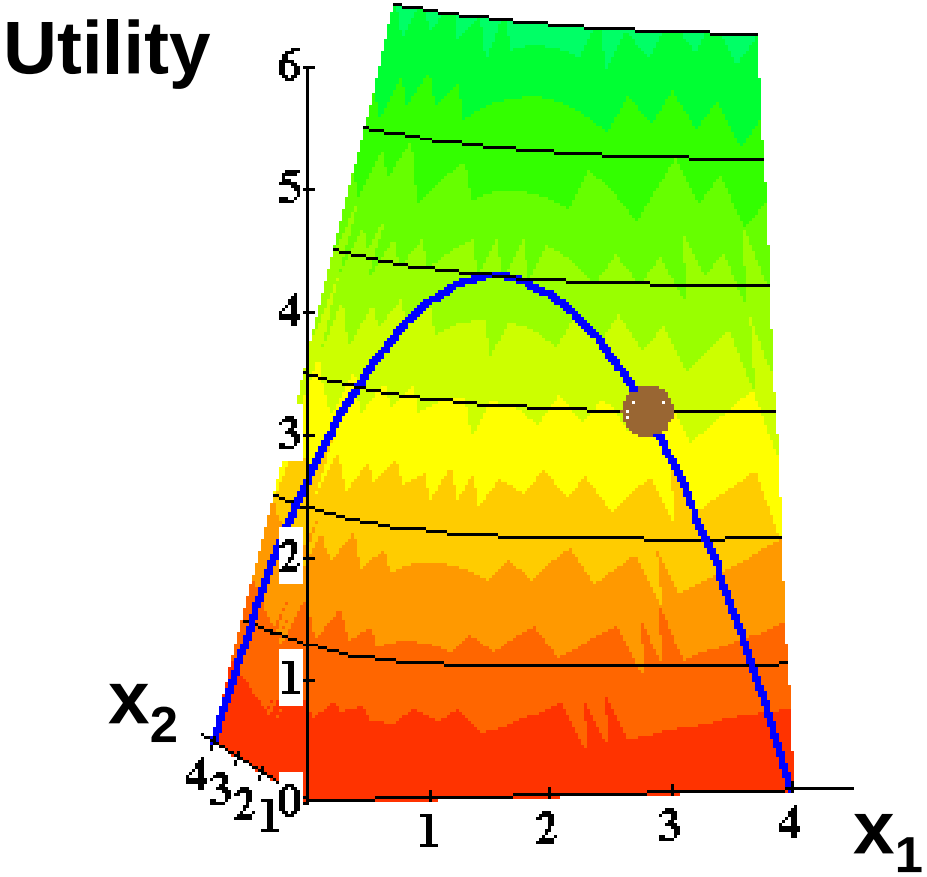

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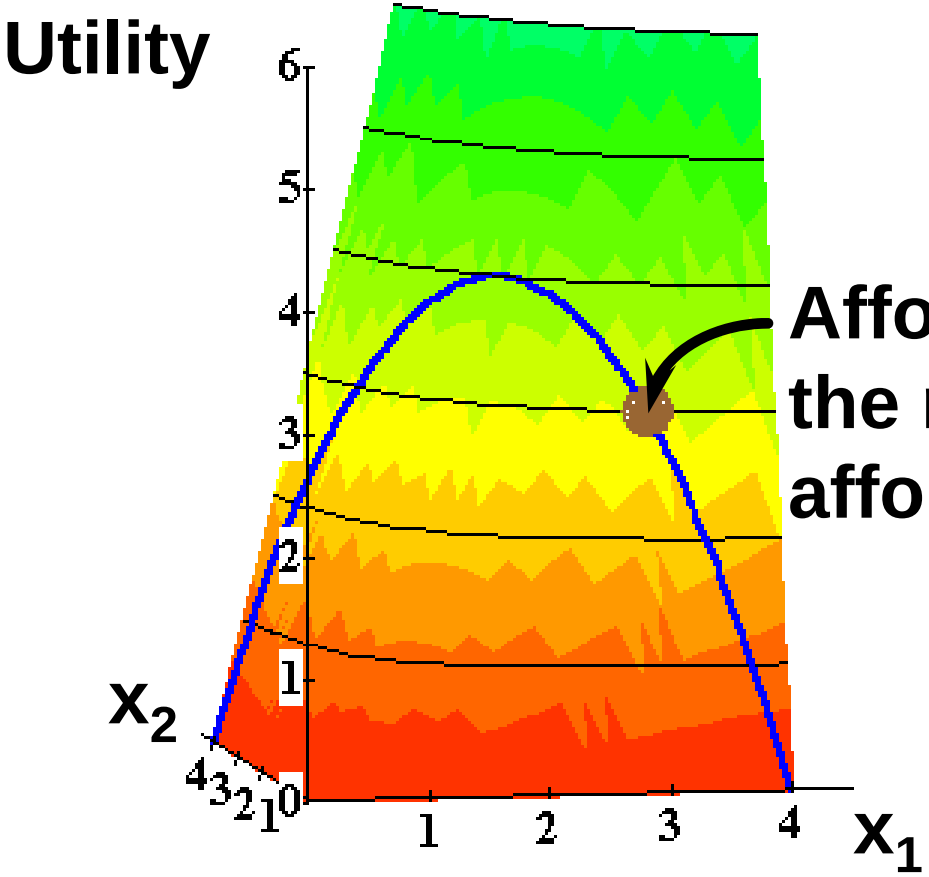
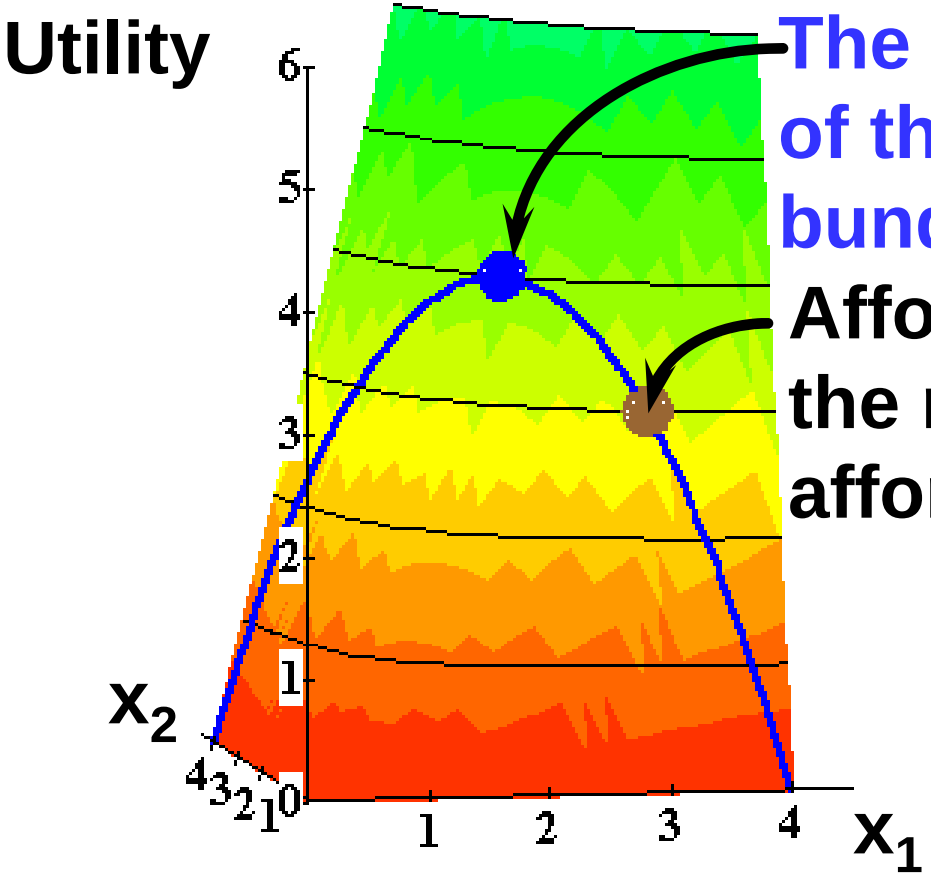



**Affordable, but not the most preferred affordable bundle.**

# Rational Constrained Choice: An Example




**The most preferred of the affordable bundles.**

**Affordable, but not the most preferred affordable bundle.**

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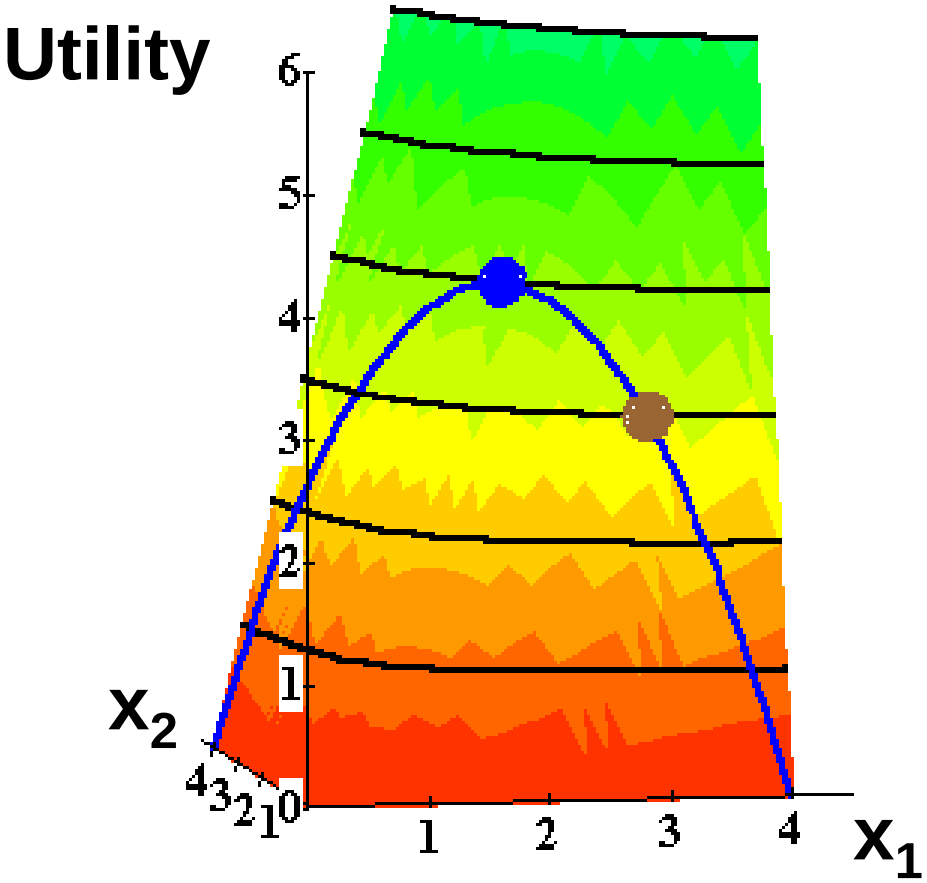

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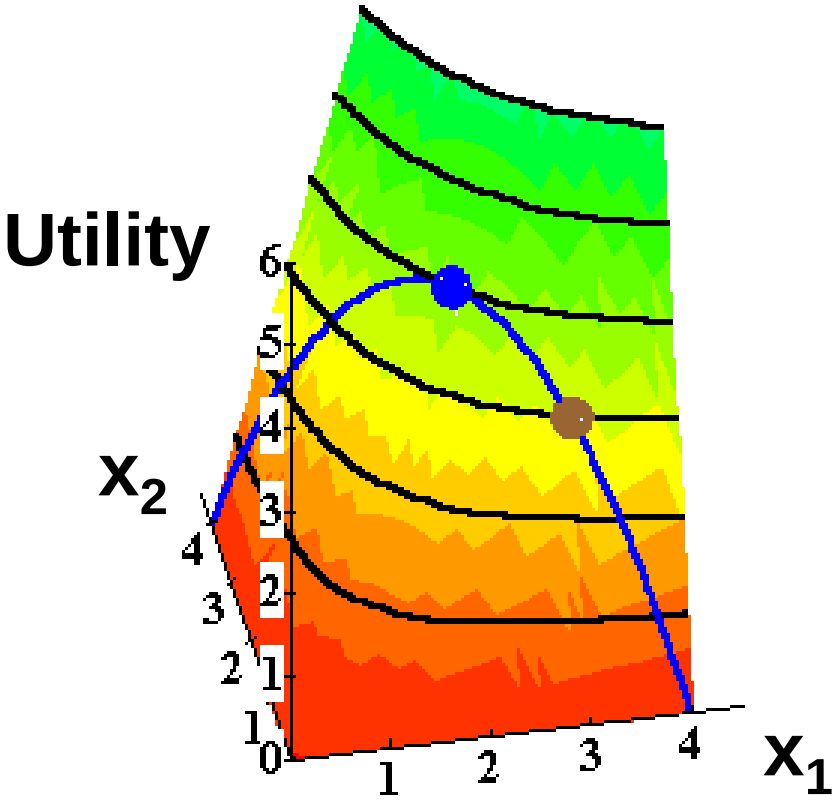

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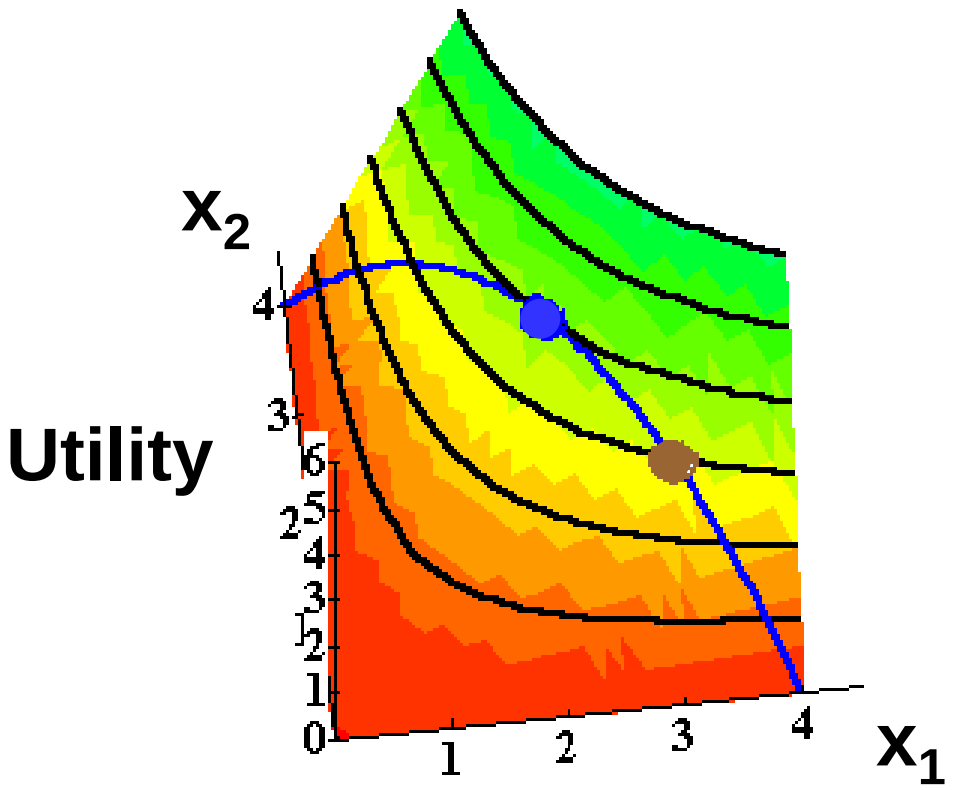

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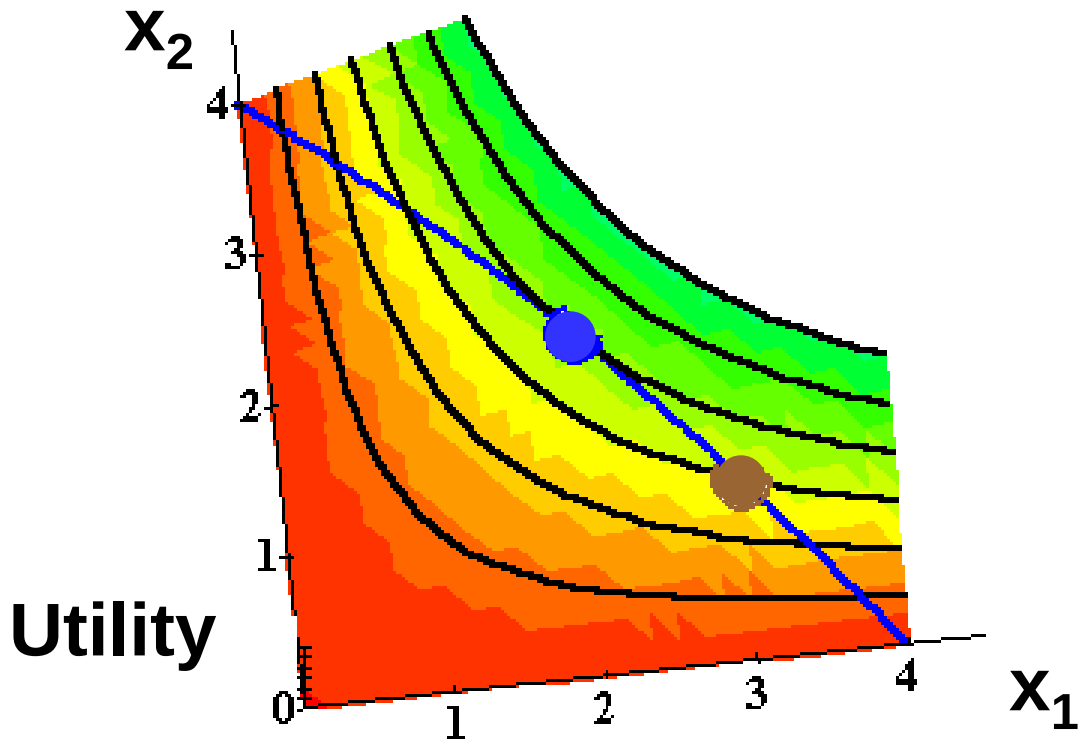

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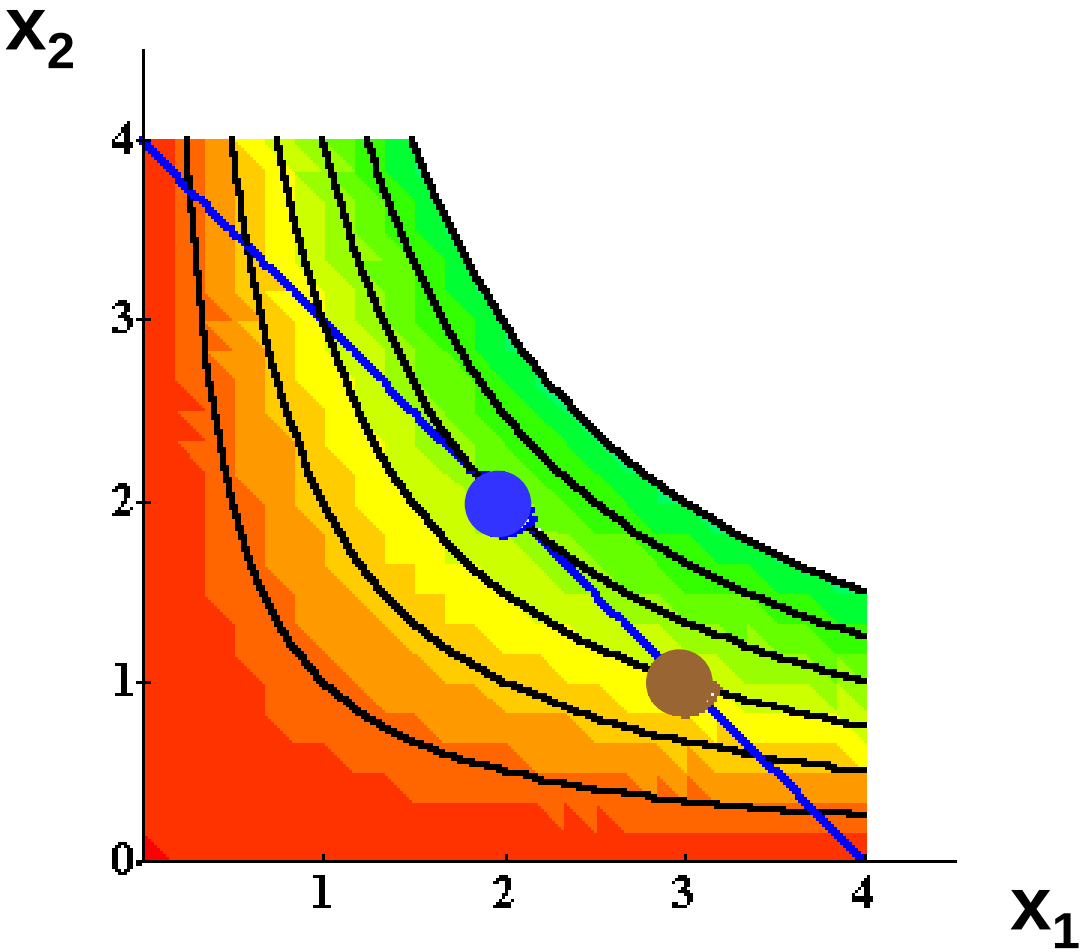

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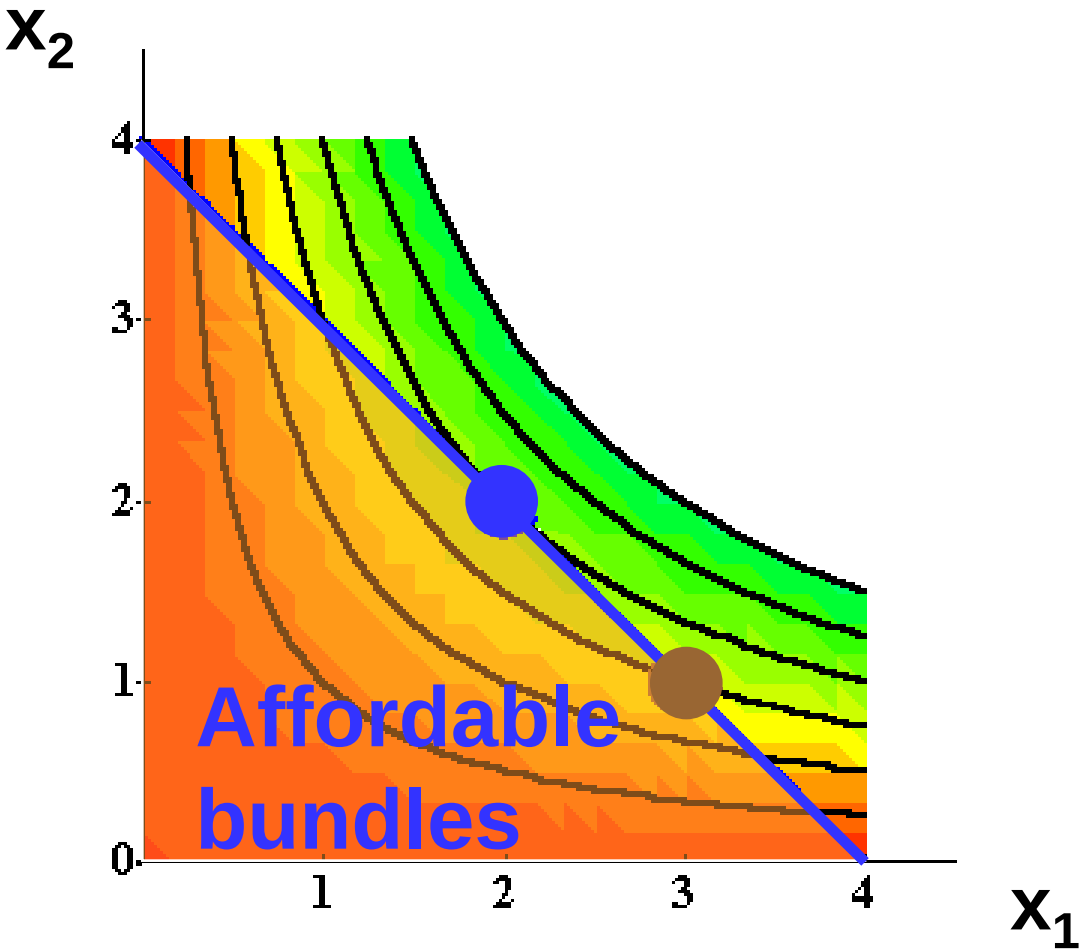

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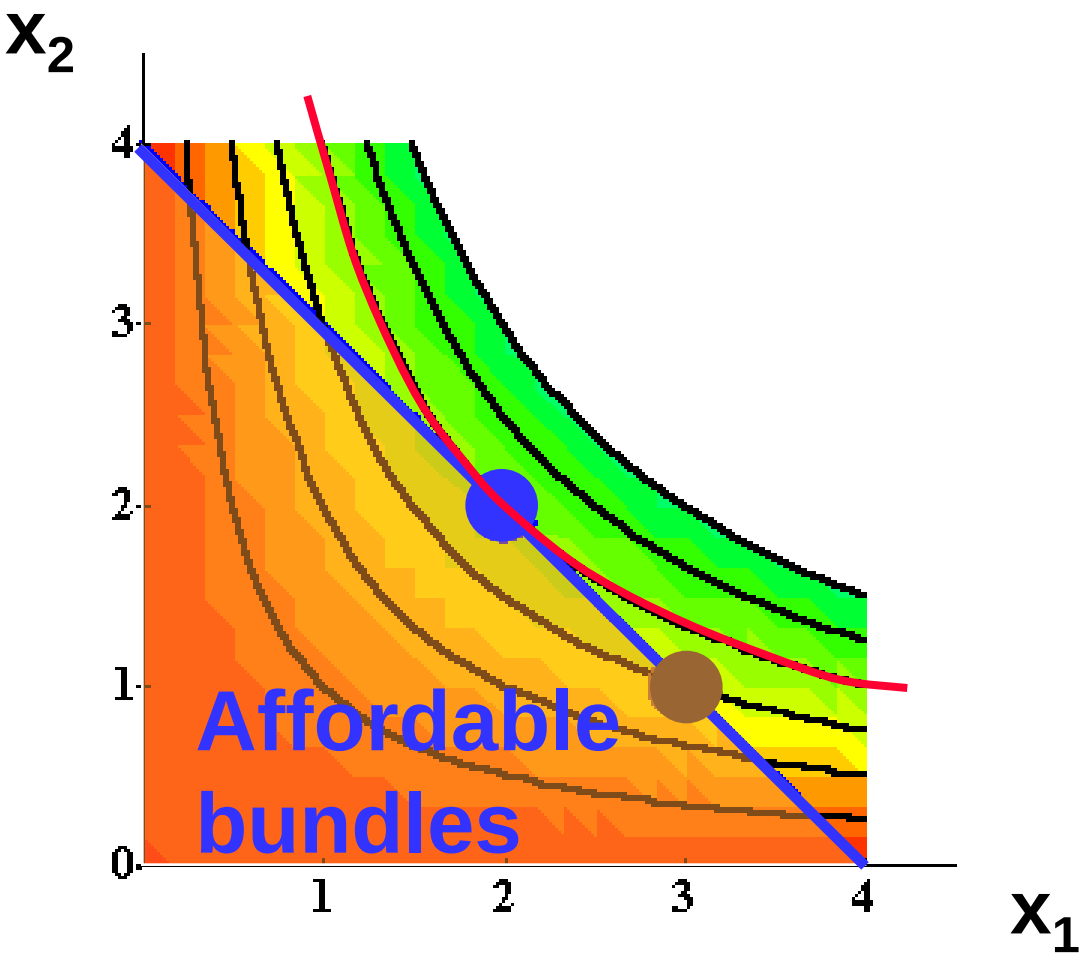

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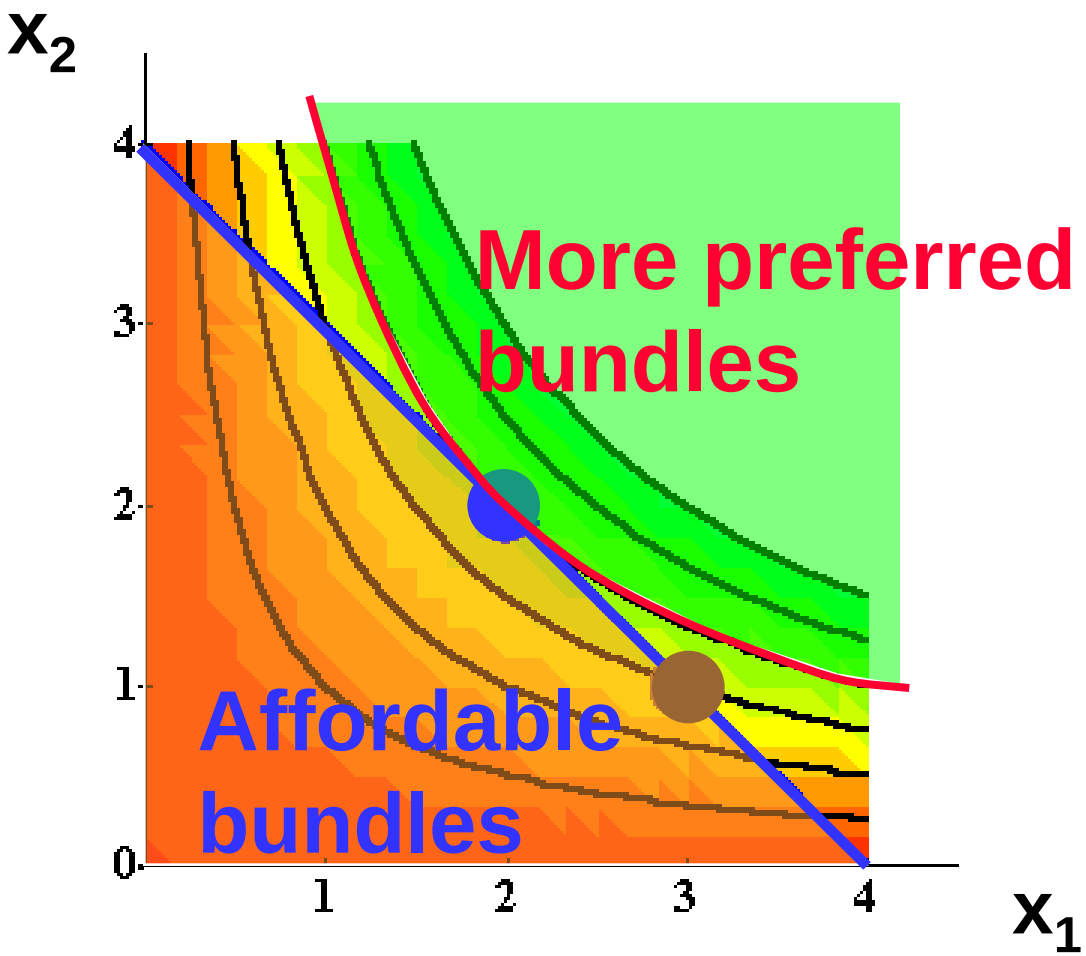

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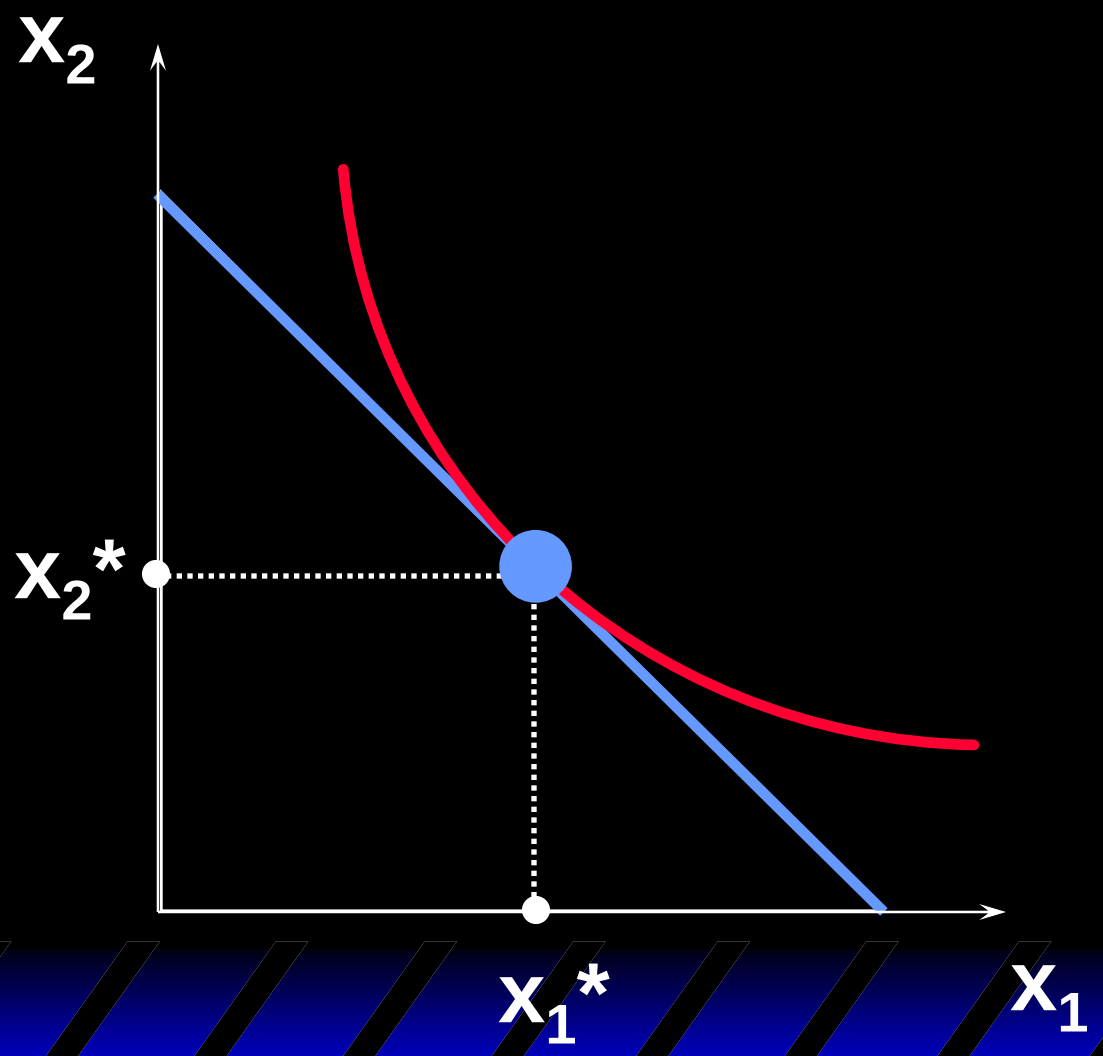

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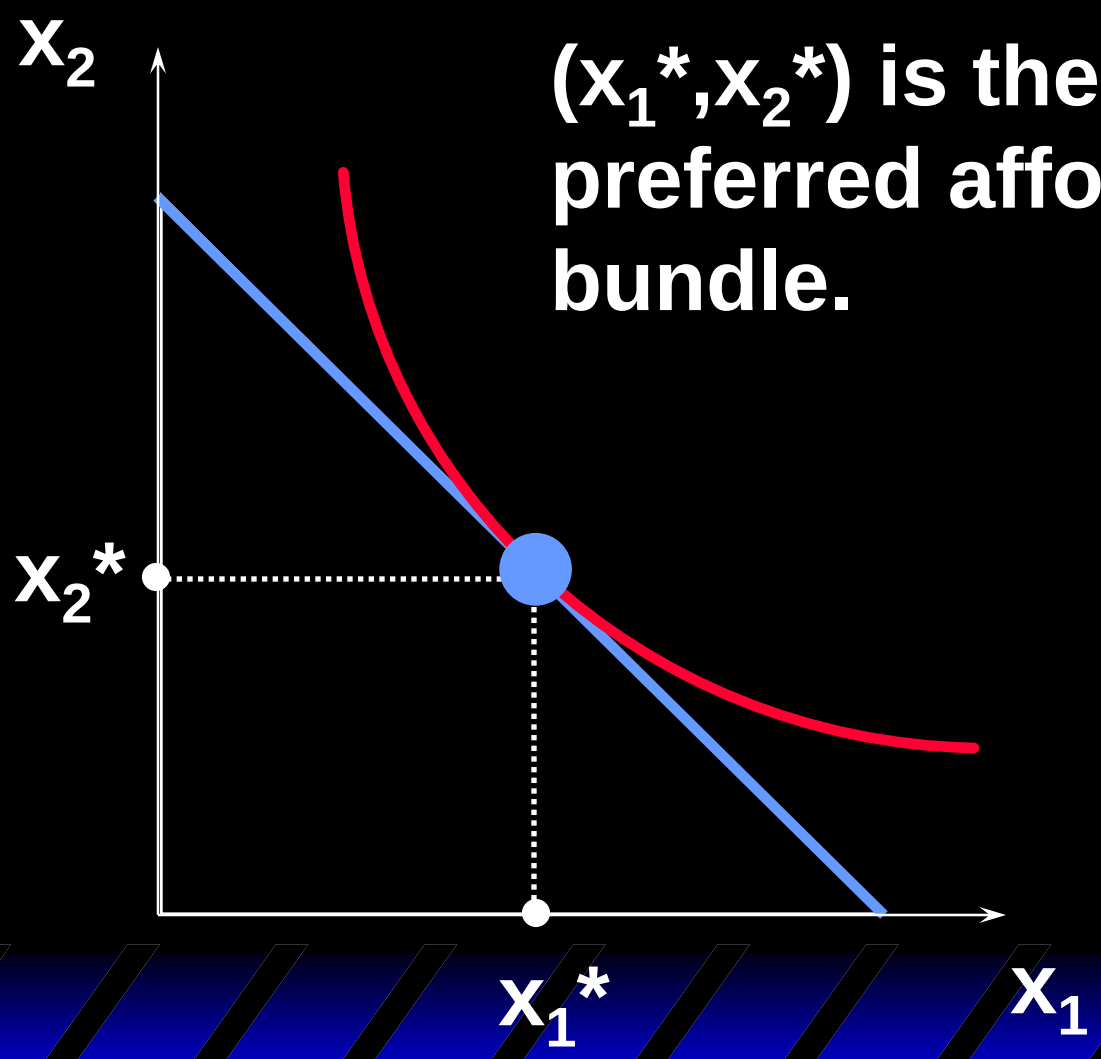



**$(x_1^*, x_2^*)$ is the most preferred affordable bundle.**

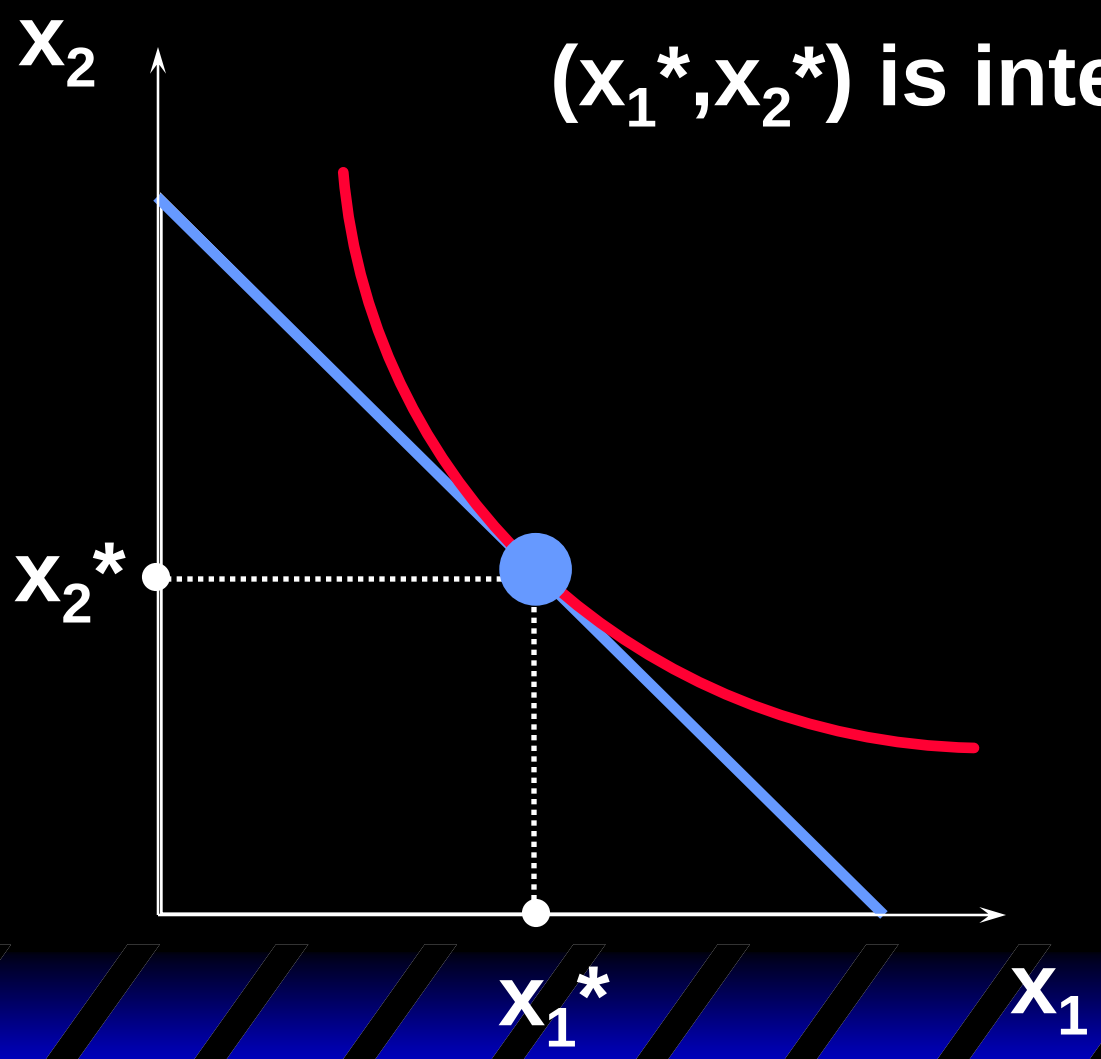
# Rational Constrained Choice

- **The most preferred affordable bundle is called the consumer's ORDINARY DEMAND at the given prices and budget.**
- **Ordinary demands will be denoted by $x_1^*(p_1,p_2,m)$ and $x_2^*(p_1,p_2,m)$.**

# Rational Constrained Choice

- **When $x_1^* > 0$ and $x_2^* > 0$ the demanded bundle is INTERIOR.**
- **If buying $(x_1^*, x_2^*)$ costs \$m then the budget is exhausted.**

# Rational Constrained Choice




**$(x_1^*, x_2^*)$ is interior.**

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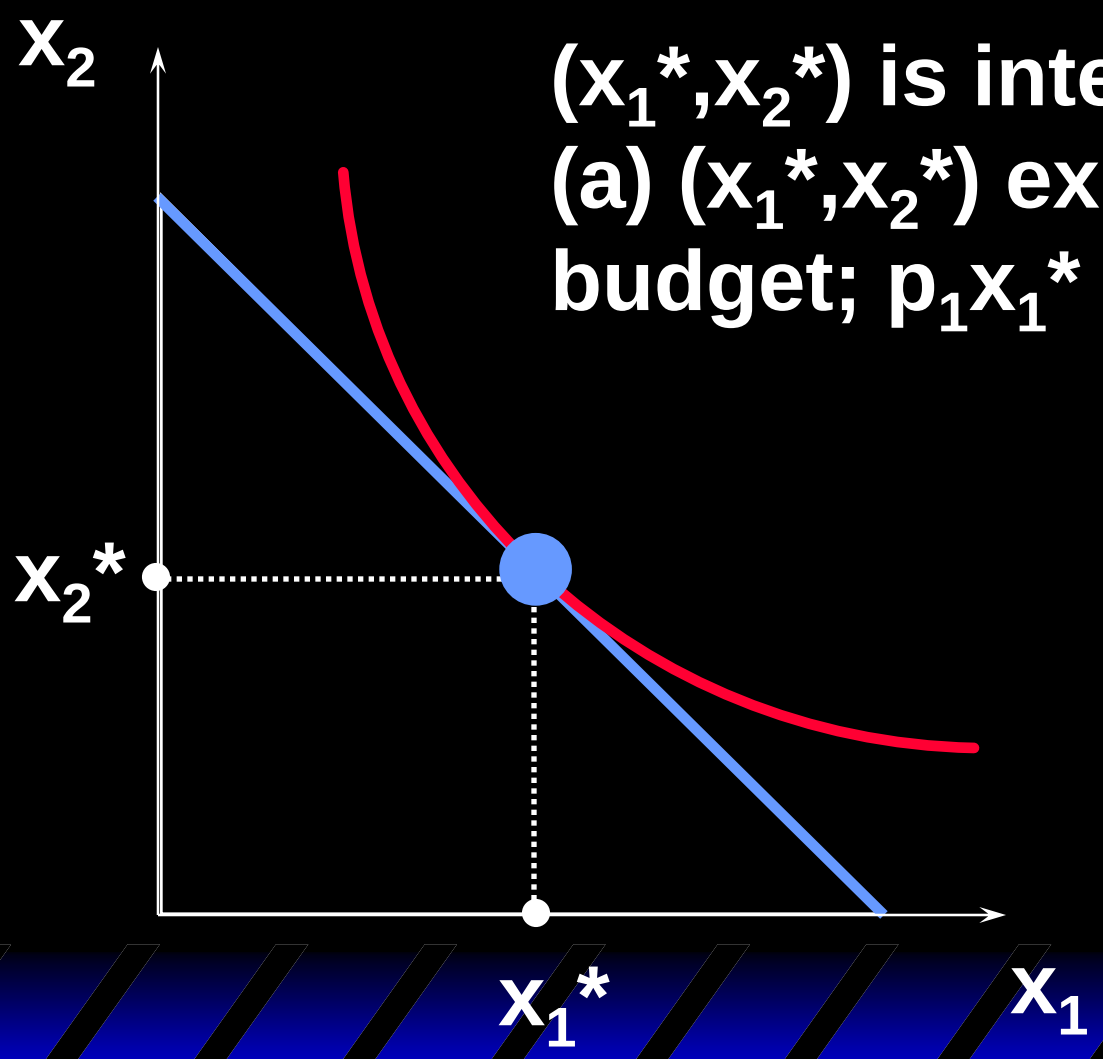
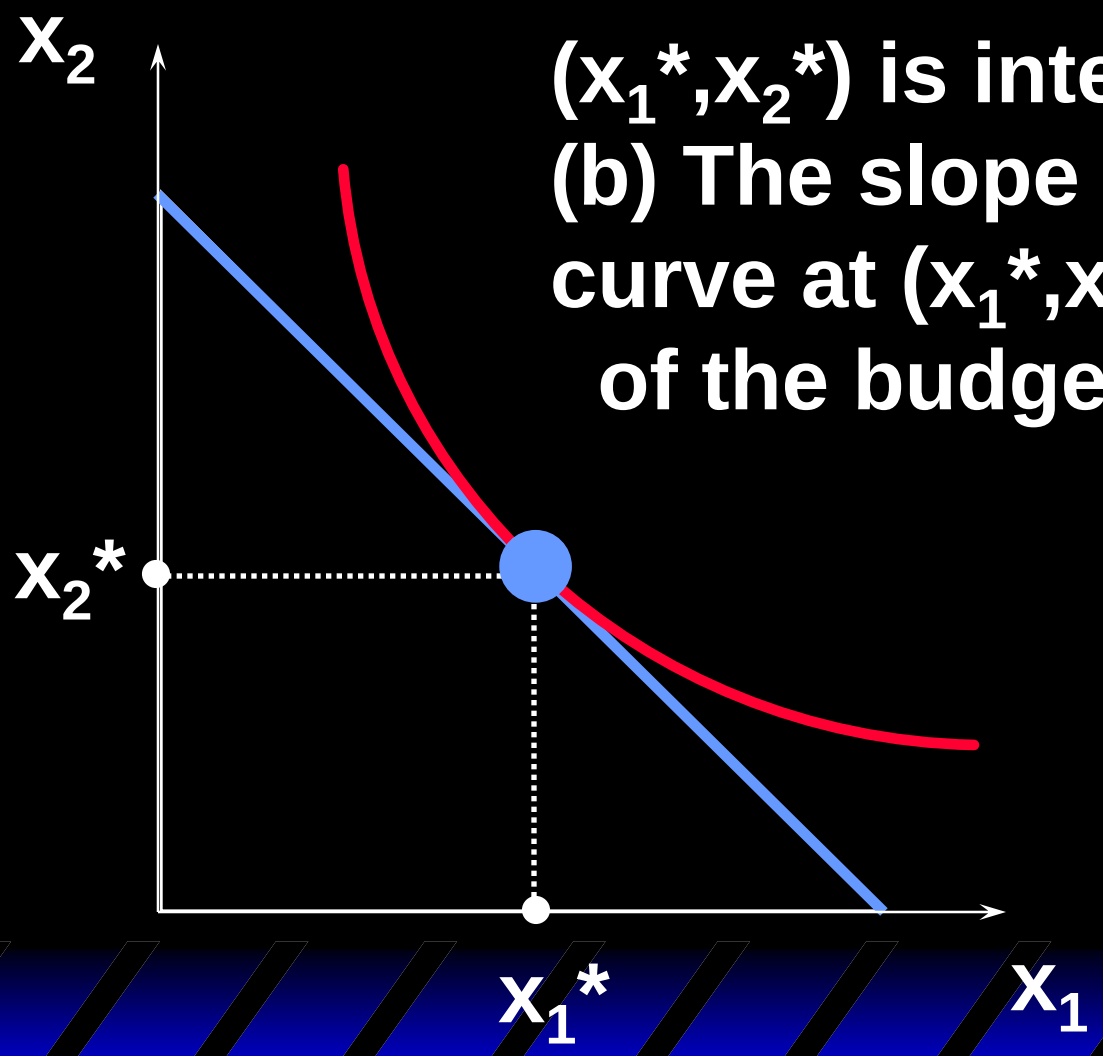



**$(x_1^*, x_2^*)$ is interior.**
**(a) $(x_1^*, x_2^*)$ exhausts the budget; $p_1x_1^* + p_2x_2^* = m$.**

# Rational Constrained Choice




**$(x_1^*,x_2^*)$ is interior .**
**(b) The slope of the indiff. curve at $(x_1^*,x_2^*)$ = slope of the budget constraint.**

# Computing Ordinary Demands

- **How can this information be used to locate $(x_1^*, x_2^*)$ for given $p_1$, $p_2$ and m?**
- **At $(x_1^*, x_2^*)$, MRS = $-p_1/p_2$**

# Computing Ordinary Demands - a Cobb-Douglas Example.

- **Suppose that the consumer has Cobb-Douglas preferences.**

$$U(x_1, x_2) = x_1^a x_2^b$$

# Computing Ordinary Demands - a Cobb-Douglas Example.

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$$U(x_1, x_2) = x_1^a x_2^b$$

- **Then**

$$MU_1 = \frac{\partial U}{\partial x_1} = a x_1^{a-1} x_2^b$$

$$MU_2 = \frac{\partial U}{\partial x_2} = b x_1^a x_2^{b-1}$$

# Computing Ordinary Demands - a Cobb-Douglas Example.

- **So the MRS is**

$$\mathbf{MRS} = \frac{dx_2}{dx_1} = -\frac{\partial U/\partial x_1}{\partial U/\partial x_2} = -\frac{ax_1^{a-1}x_2^{b}}{bx_1^{a}x_2^{b-1}} = -\frac{ax_2}{bx_1}.$$

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- **At $(x_1^*, x_2^*)$, MRS = $-p_1/p_2$ so**

$$-\frac{ax_2^*}{bx_1^*} = -\frac{p_1}{p_2} \quad \Rightarrow \quad x_2^* = \frac{bp_1}{ap_2}x_1^*. \qquad \textbf{(A)}$$

# Computing Ordinary Demands - a Cobb-Douglas Example.

- **$(x_1^*, x_2^*)$ also exhausts the budget so**

$$p_1 x_1^* + p_2 x_2^* = m. \qquad \textbf{(B)}$$

# Computing Ordinary Demands - a Cobb-Douglas Example.

- **So now we know that**

$$x_2^* = \frac{bp_1}{ap_2} x_1^* \qquad \textbf{(A)}$$

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- **So now we know that**

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**Substitute**

$$p_1 x_1^* + p_2 x_2^* = m. \qquad \textbf{(B)}$$

**and get**

$$p_1 x_1^* + p_2 \frac{bp_1}{ap_2} x_1^* = m.$$

**This simplifies to ….**

# Computing Ordinary Demands - a Cobb-Douglas Example.

$$x_1^* = \frac{am}{(a+b)p_1}.$$

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**Substituting for $x_1$* in**

$$p_1 x_1^* + p_2 x_2^* = m$$

**then gives**

$$x_2^* = \frac{bm}{(a+b)p_2}.$$

# Rational Constrained Choice

- **IF (1) the optimum is interior ($x_1^* > 0$ and $x_2^* > 0$)**

  **(2) $(x_1^*, x_2^*)$ exhausts the budget (← Strictly monotone pref),**

  **(3) Pref. are strictly convex (→ unique optimum)**

# Rational Constrained Choice

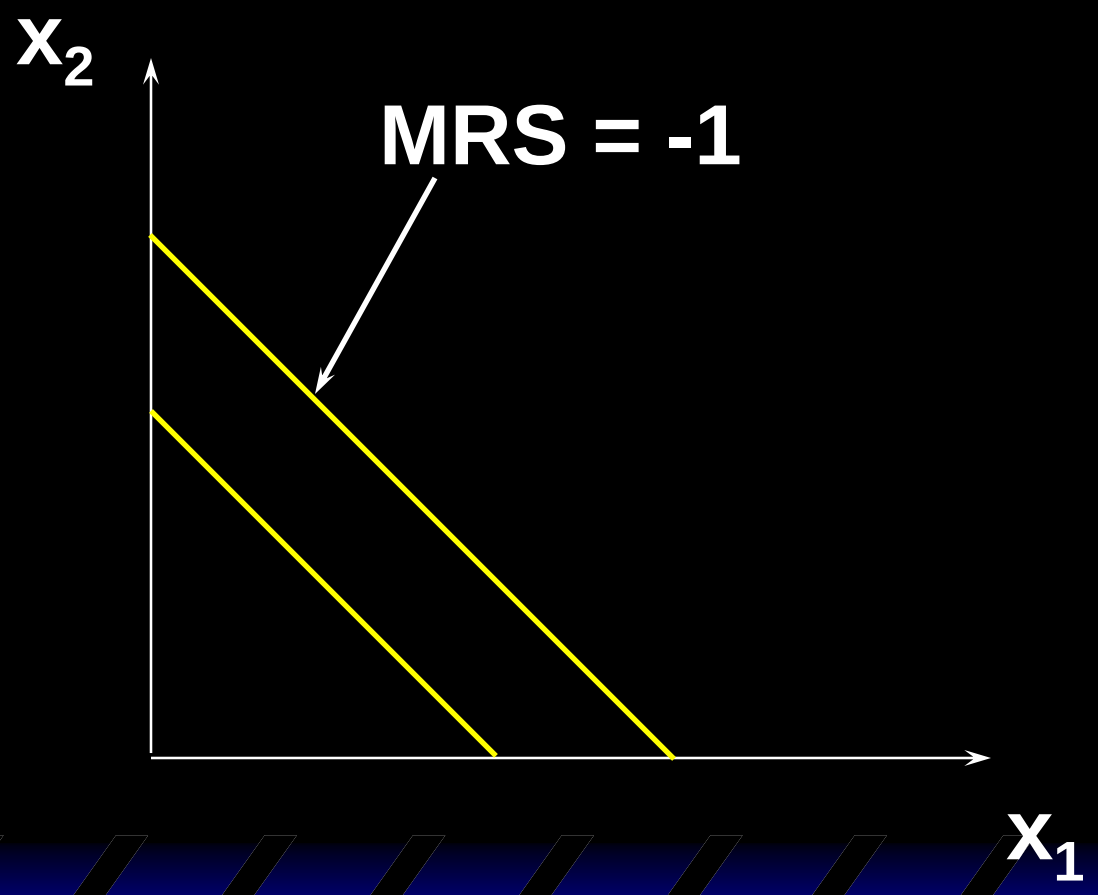
**THEN the ordinary demands are obtained by solving:**

- **(a)** $p_1x_1^* + p_2x_2^* = m$
- **(b) slope of the budget line** $-p_1/p_2$ **= slope of the indifference curve containing** $(x_1^*, x_2^*)$**.**

# Rational Constrained Choice

- **What if preferences display perfect substitution or complementarities? (A: Think)**
- **What if prices are not constant (but smooth functions)? (A: Lagrange)**

# Examples of Corner Solutions -- the Perfect Substitutes Case



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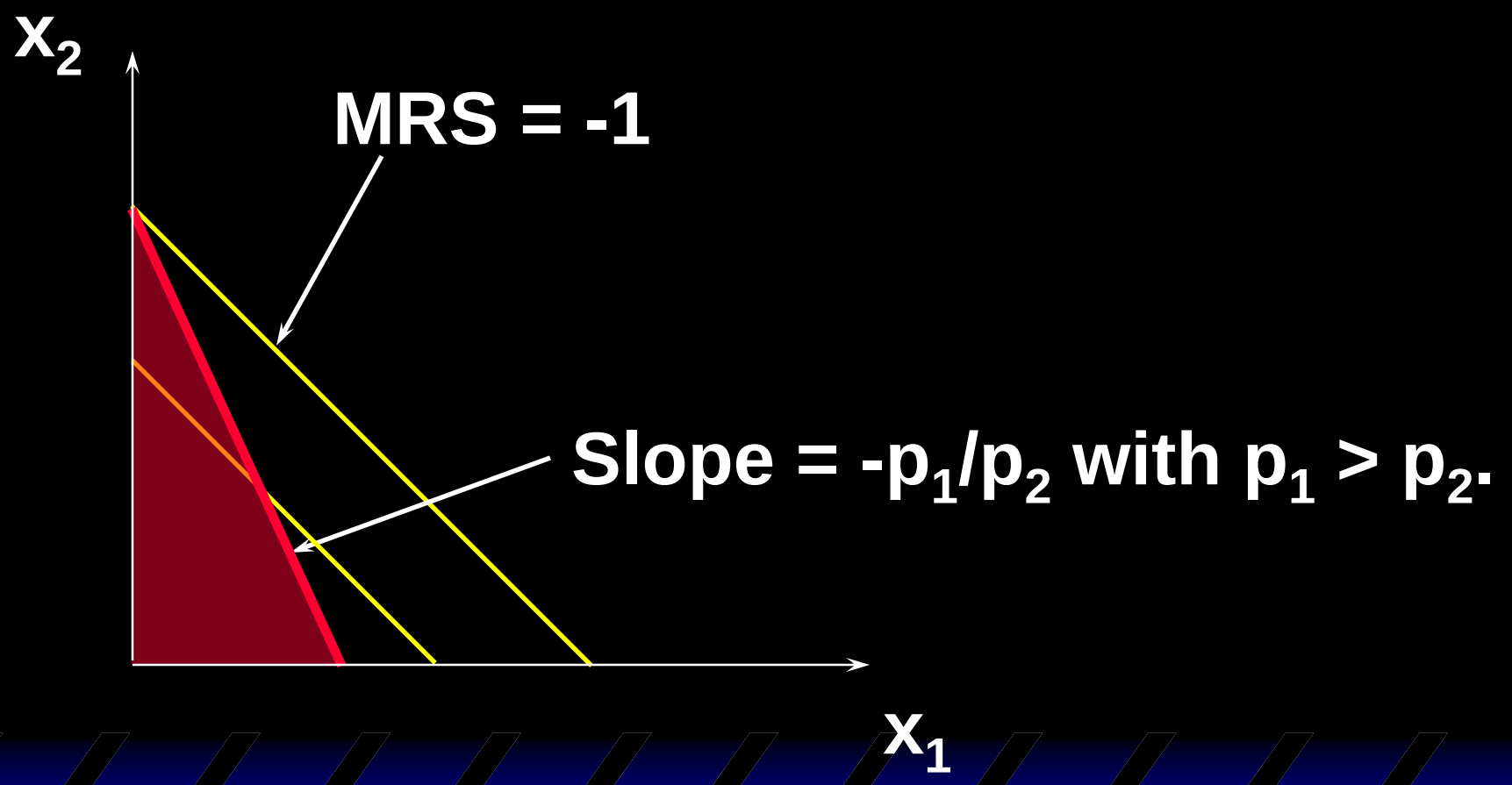

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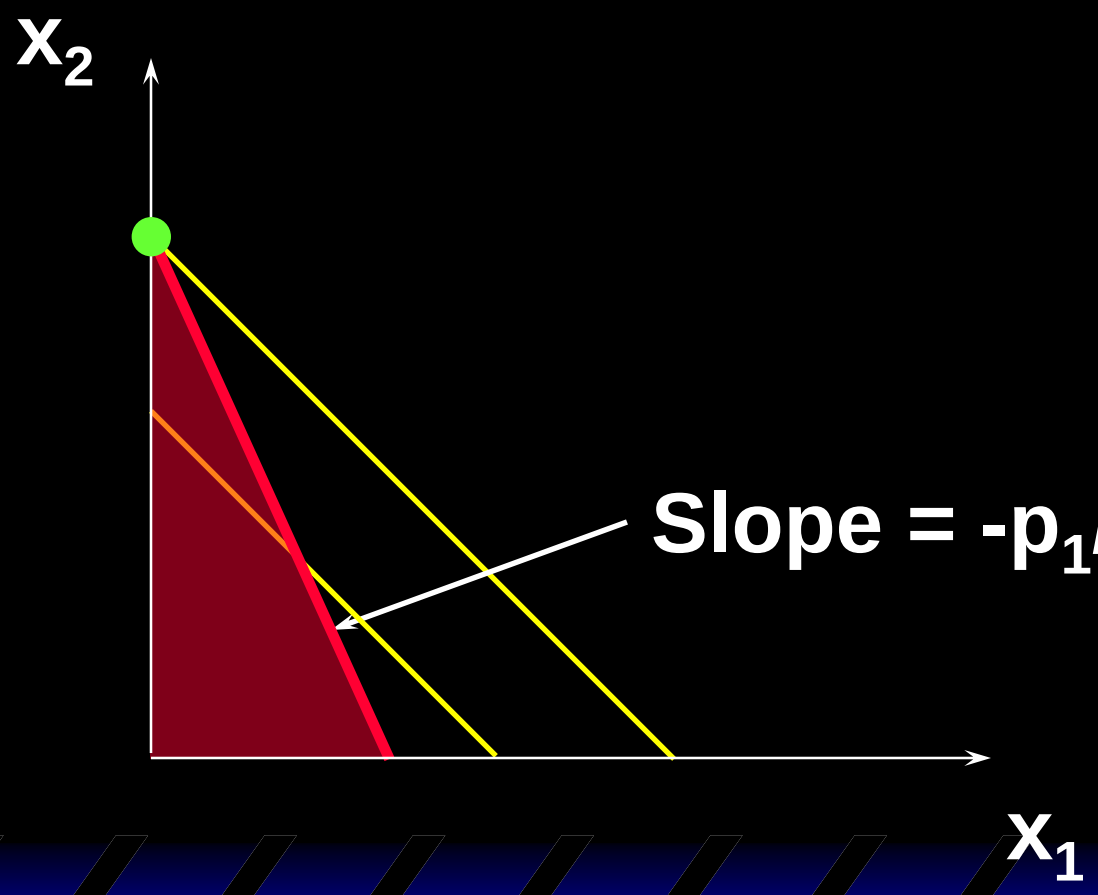


**Slope = $-p_1/p_2$ with $p_1 > p_2$.**

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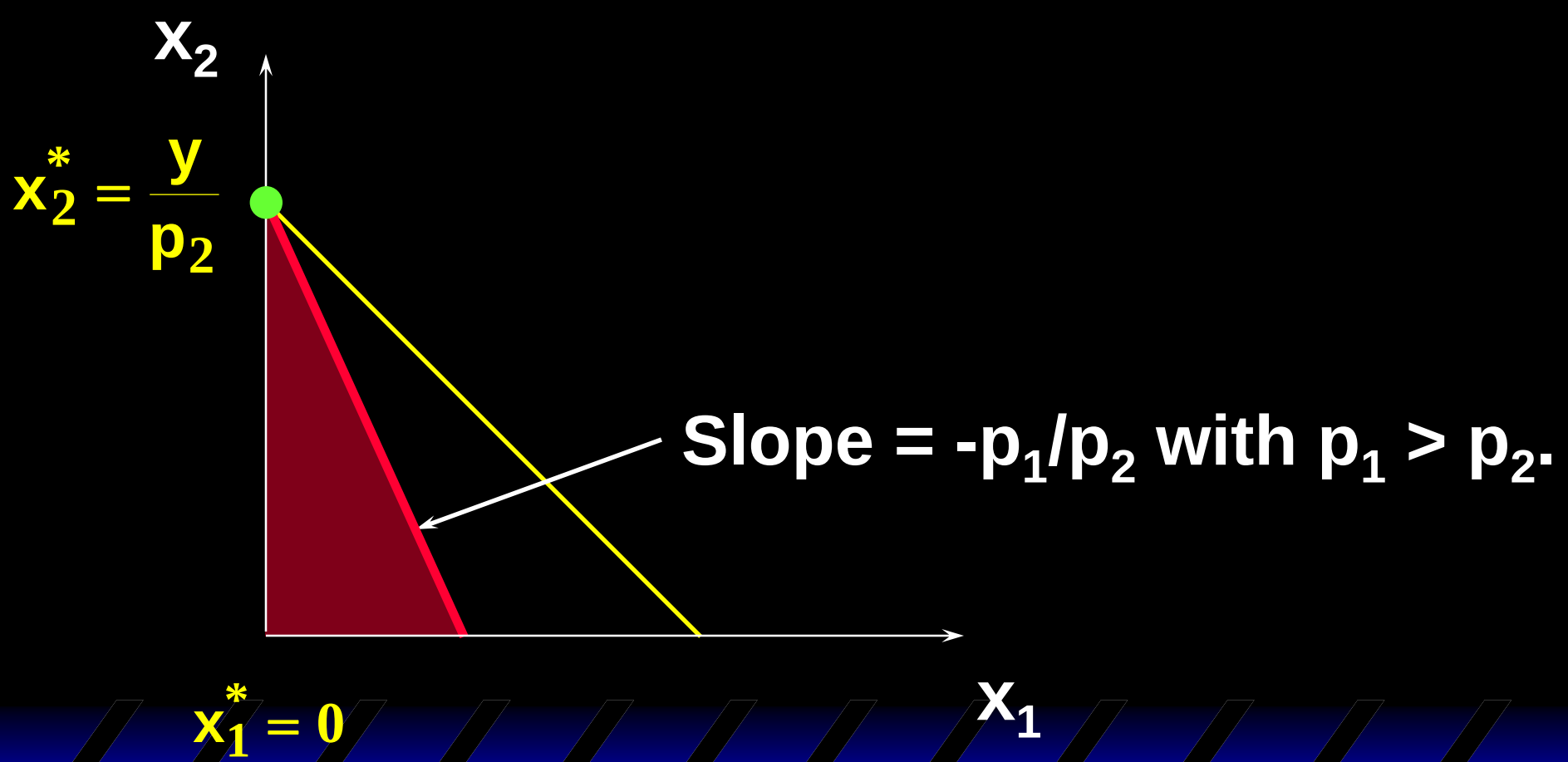

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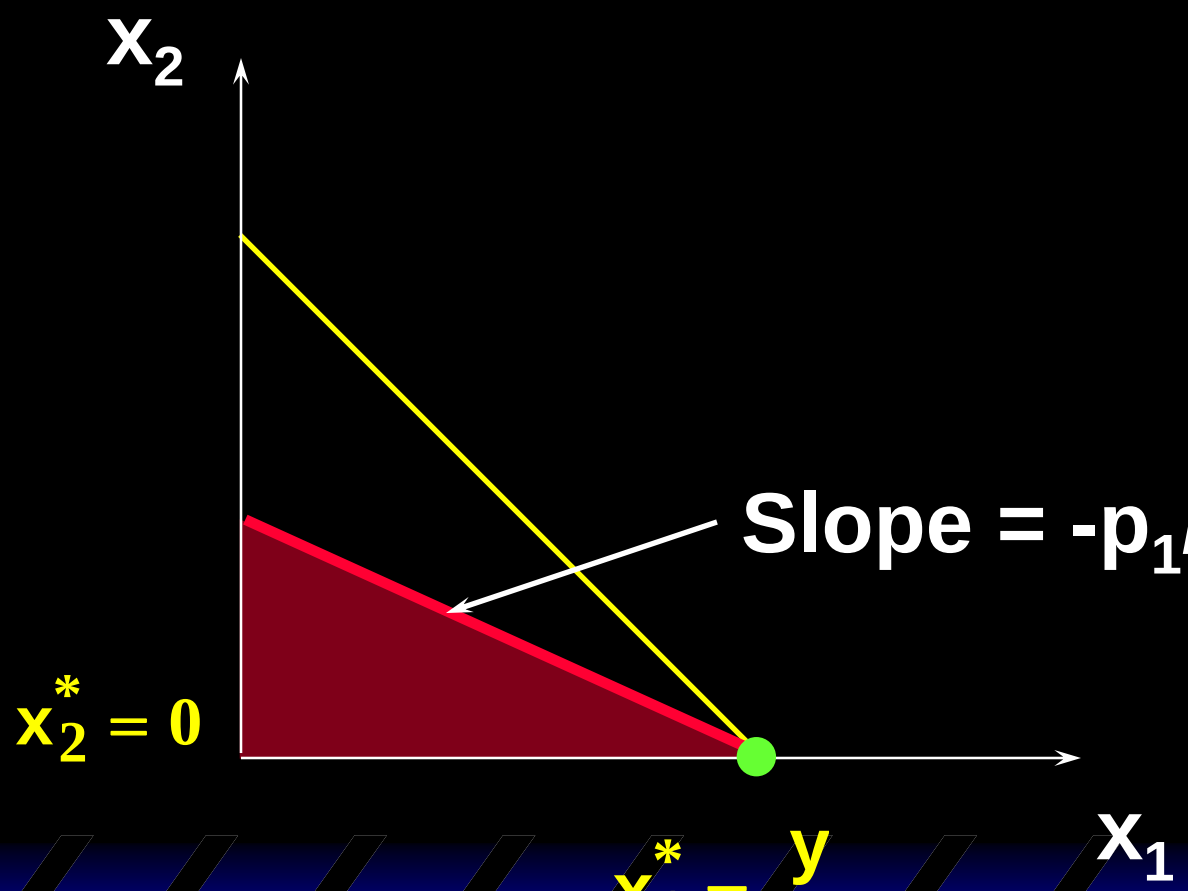

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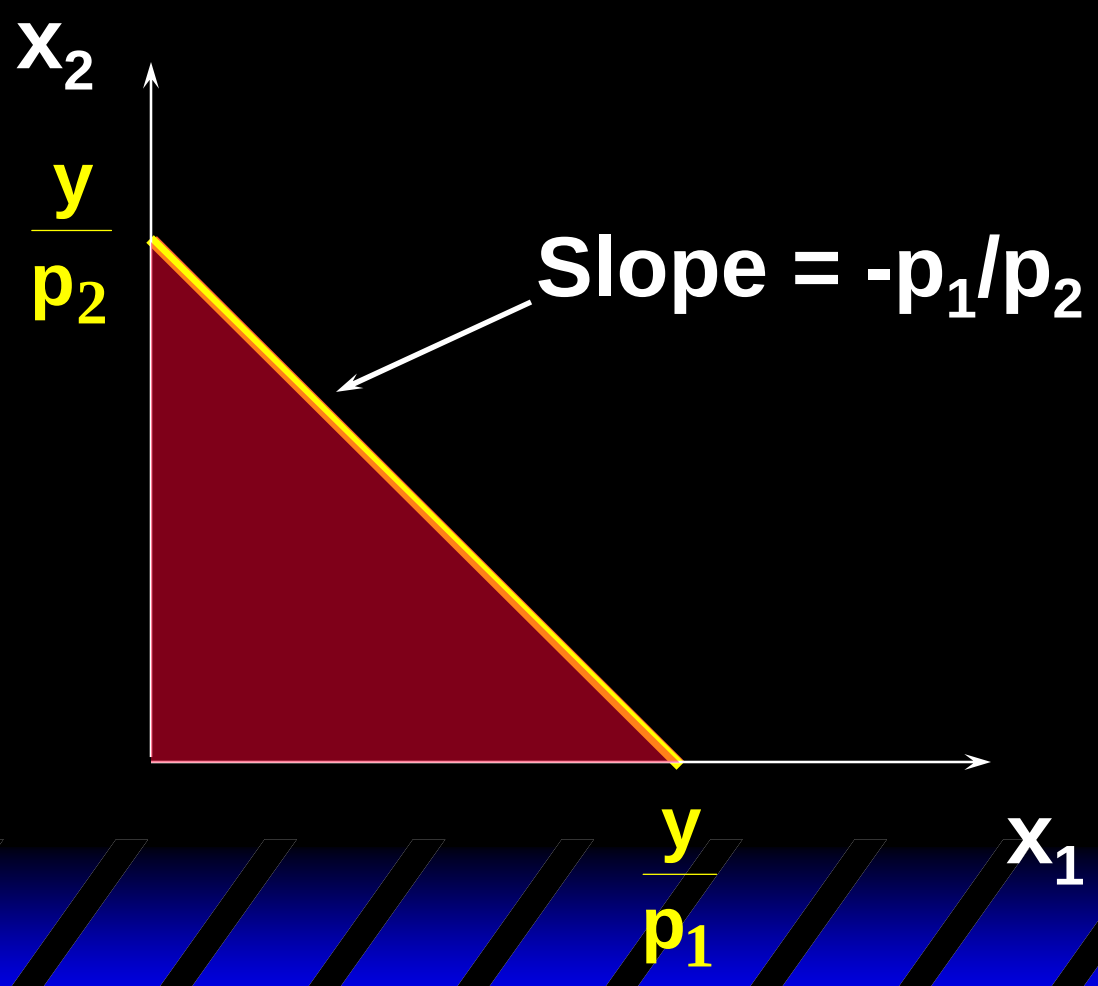
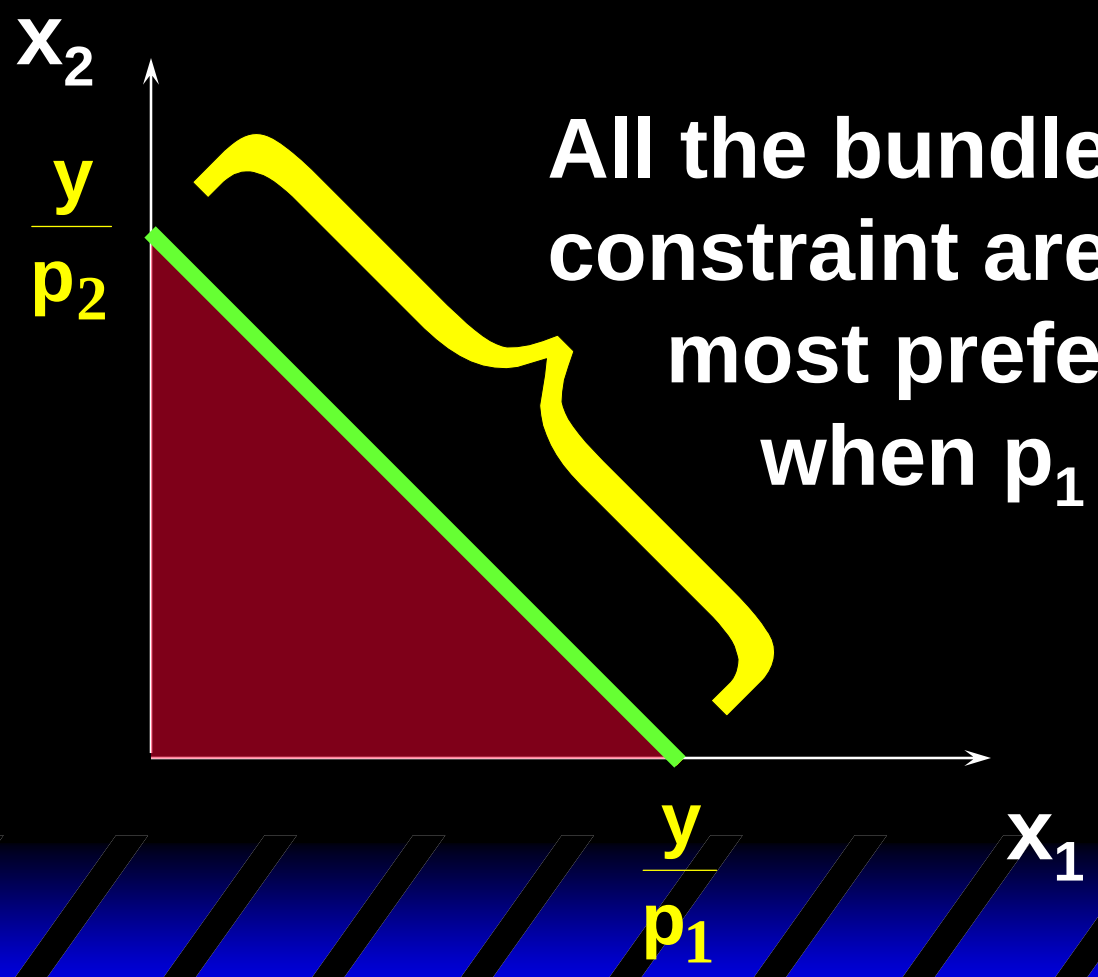


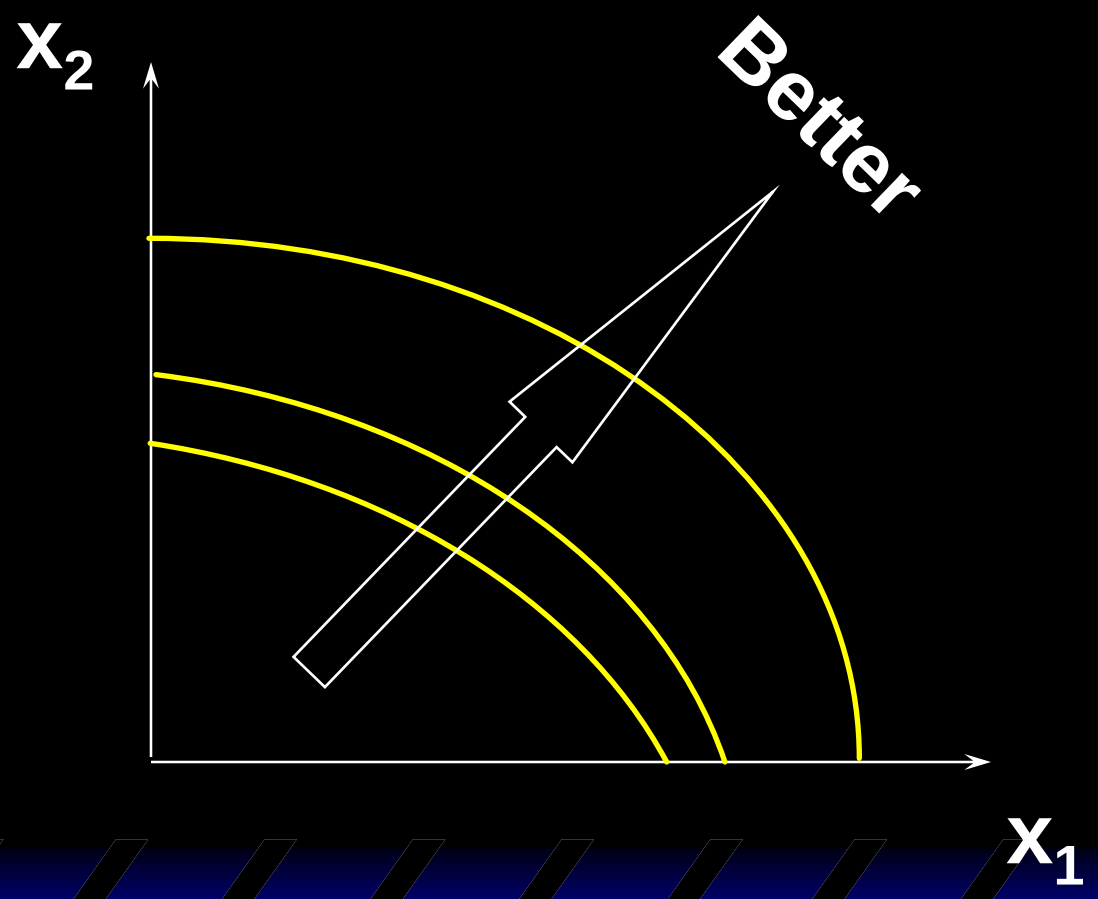

**Slope = $-p_1/p_2$ with $p_1 = p_2$.**

# Examples of Corner Solutions -- the Perfect Substitutes Case




**All the bundles in the constraint are equally the most preferred affordable when $p_1 = p_2$.**

# Examples of Corner Solutions -- the Non-Convex Preferences Case



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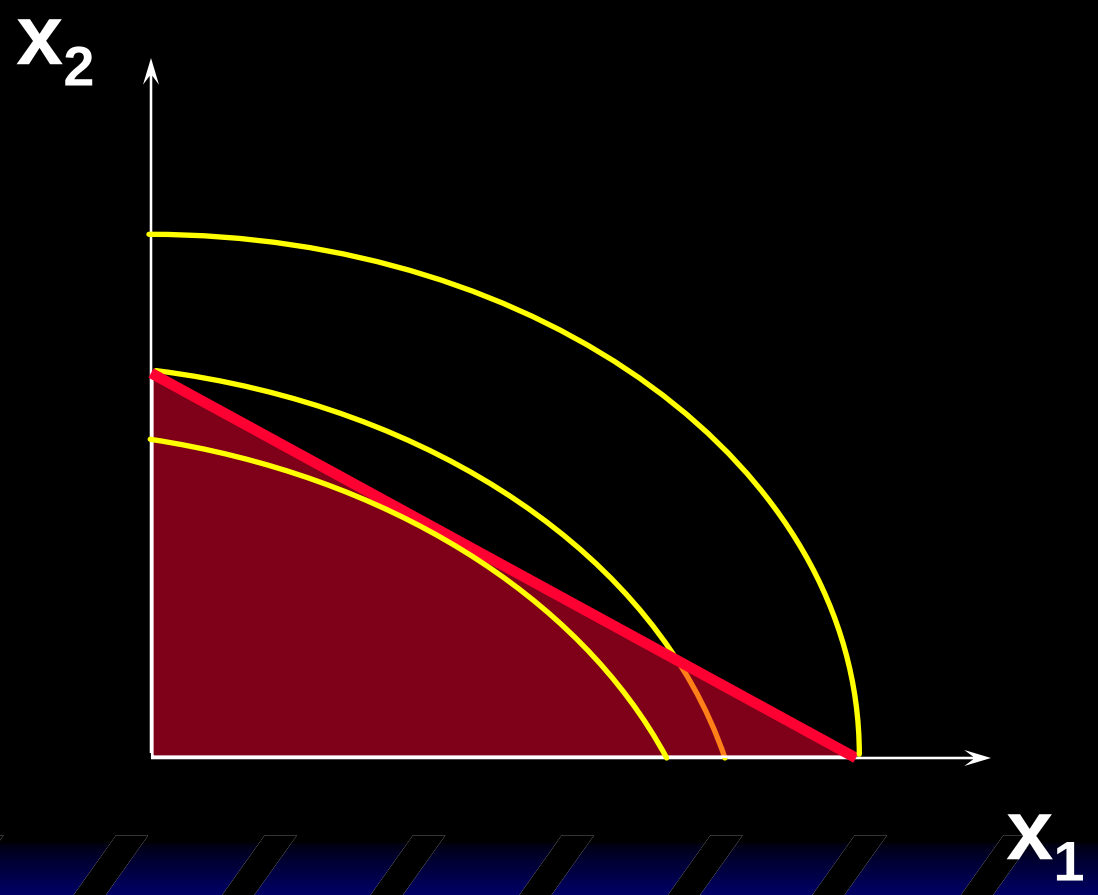

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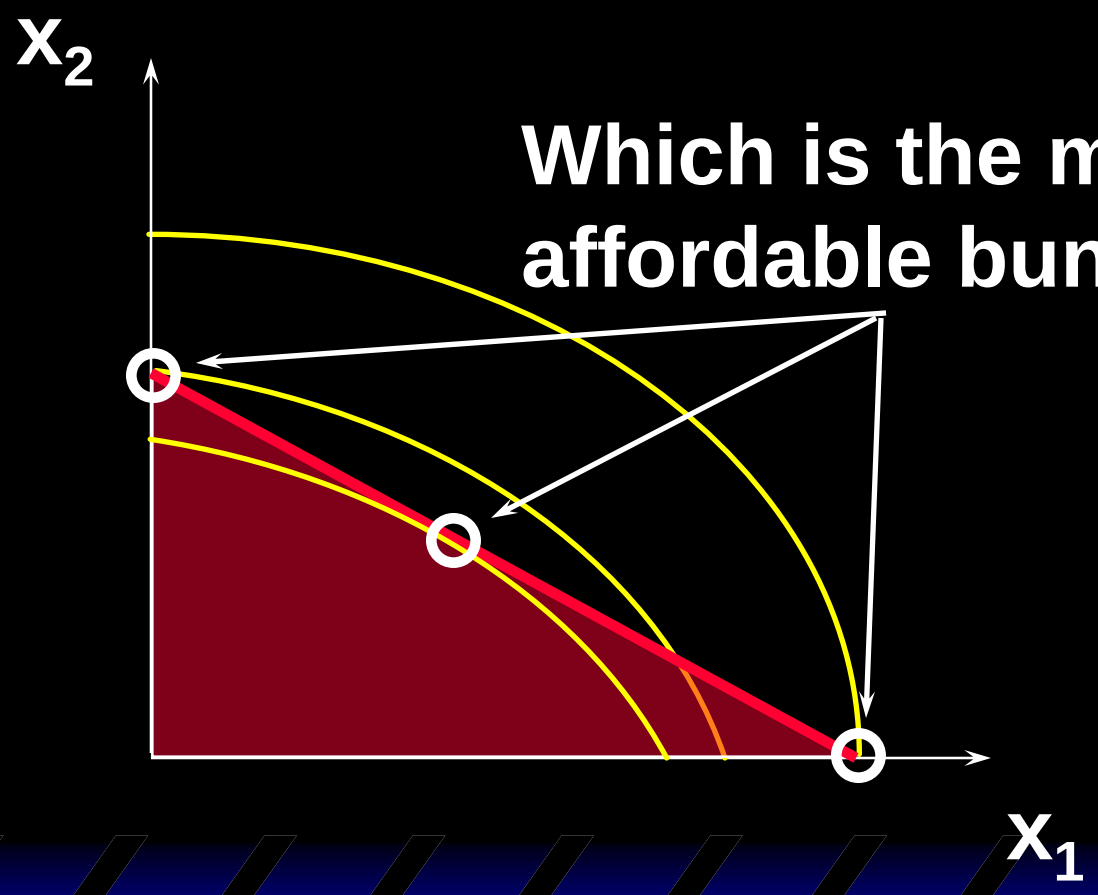


**Which is the most preferred affordable bundle?**

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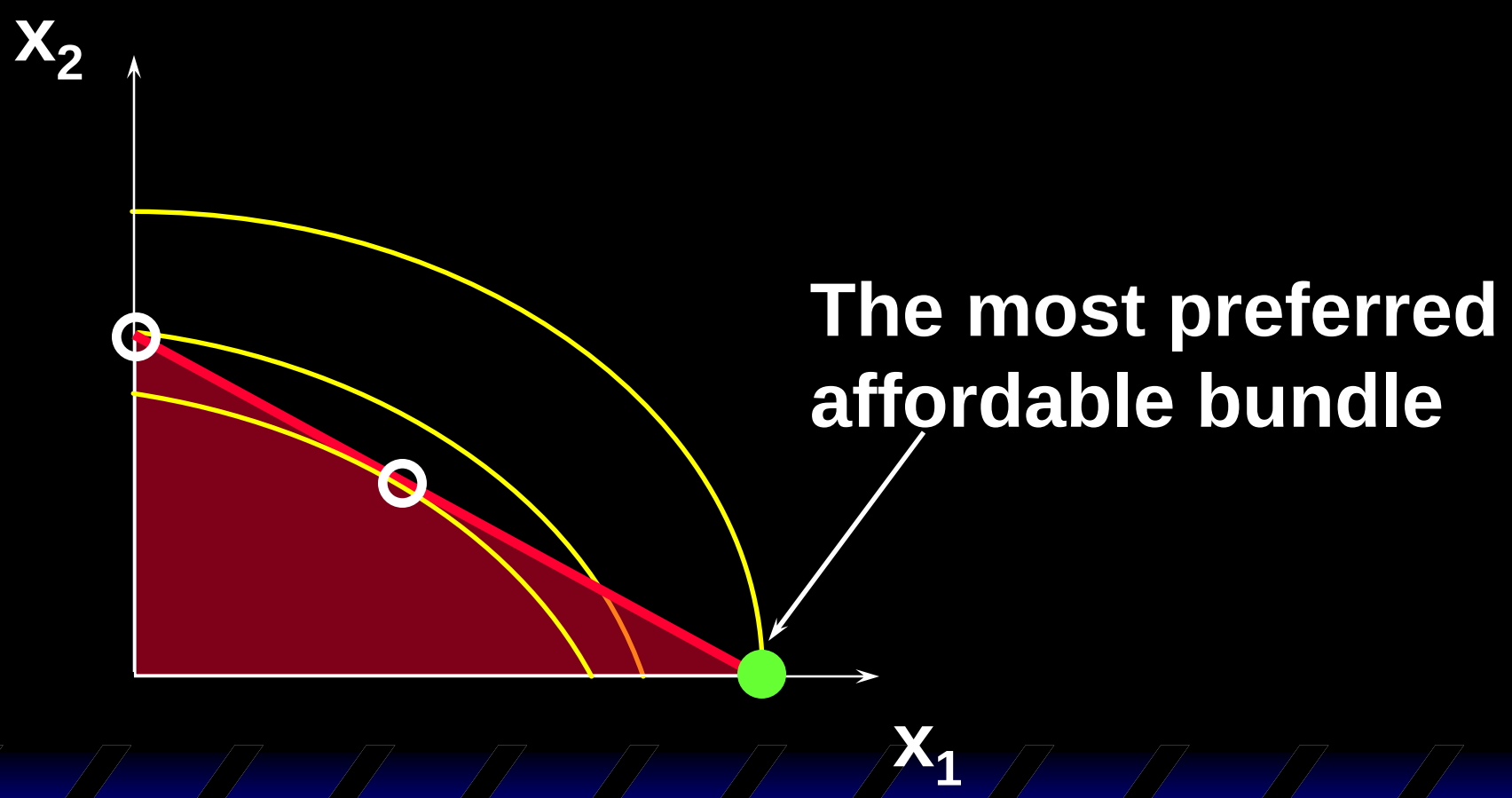

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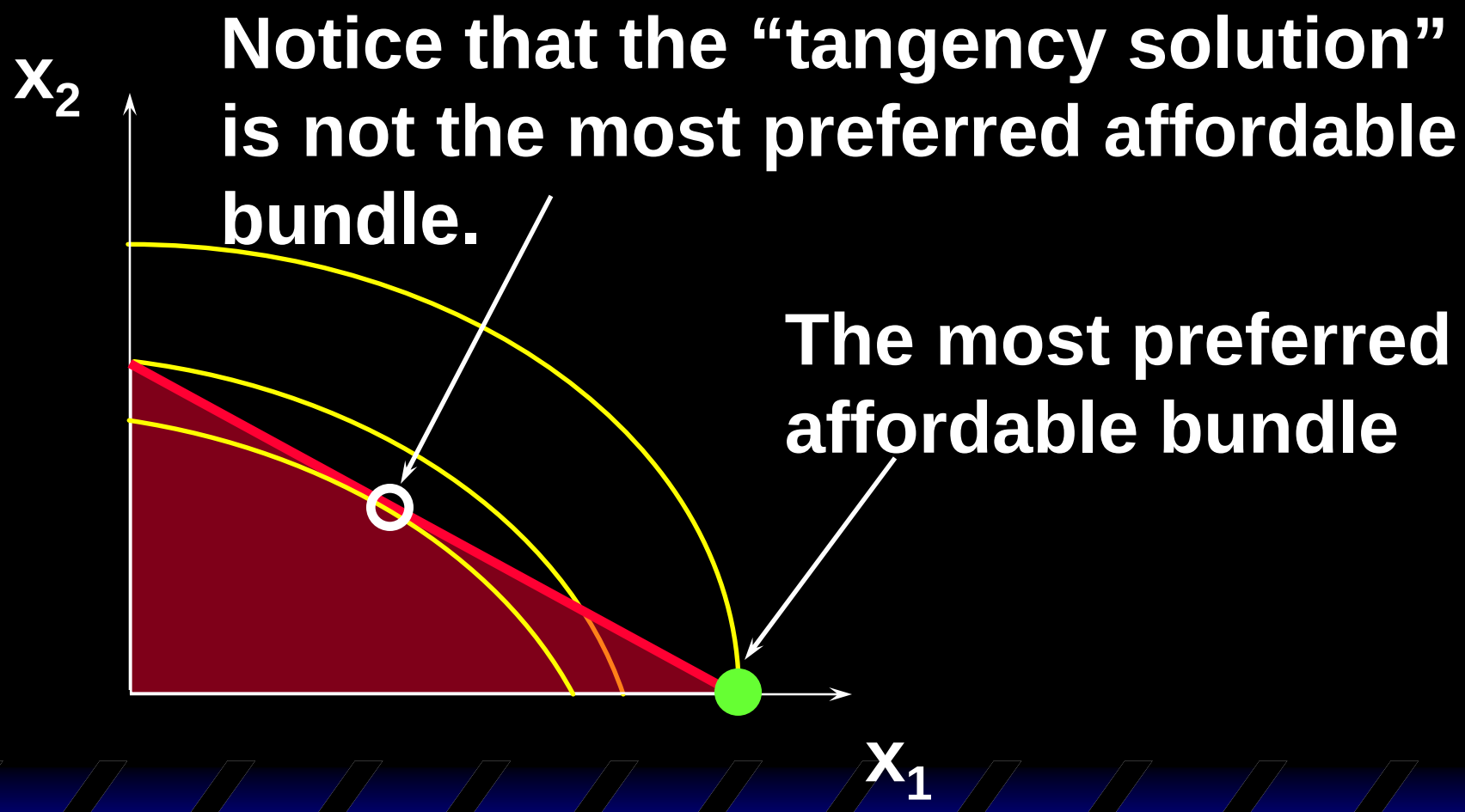
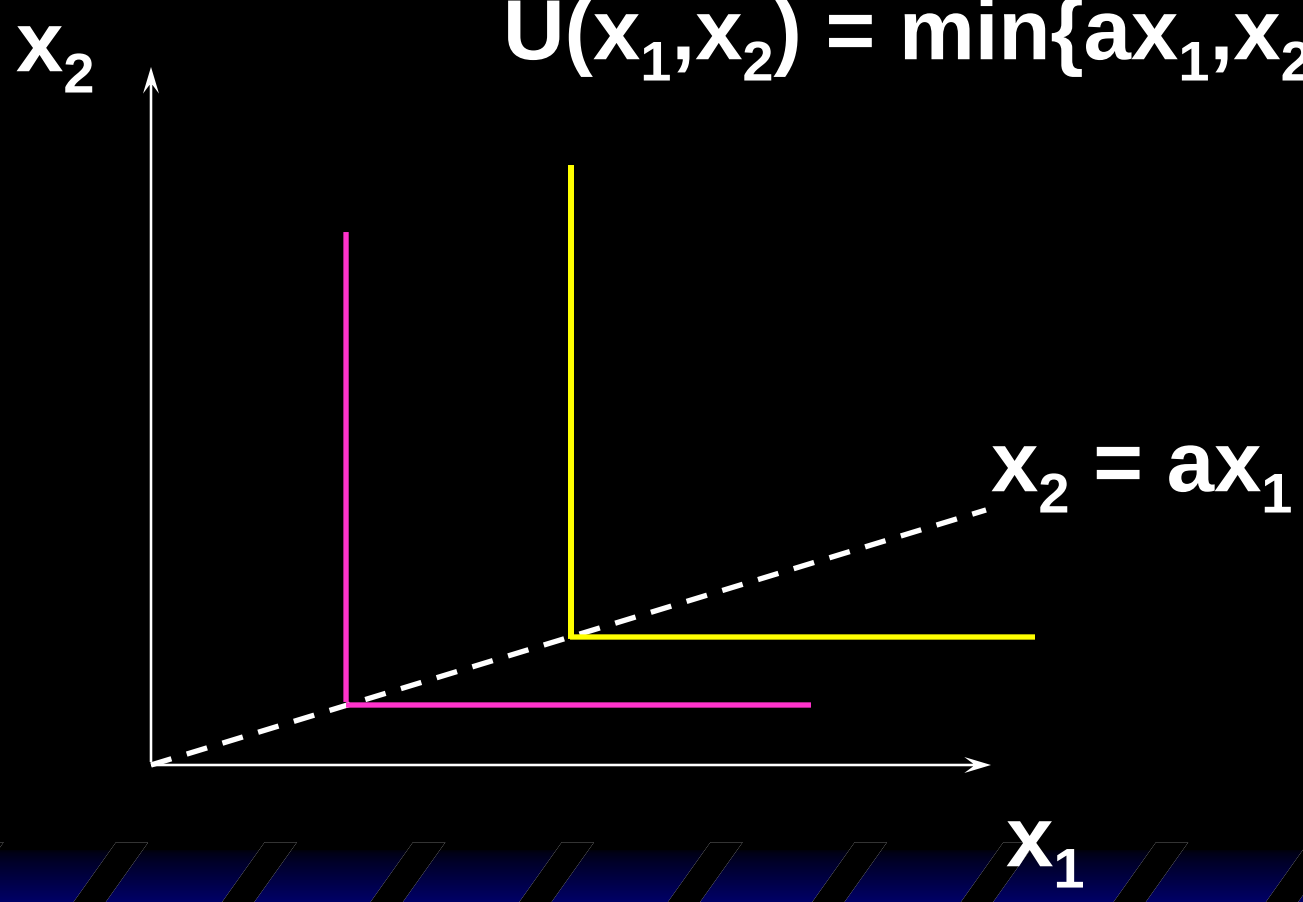


# Examples of ‘Kink’ Solutions -- the Perfect Complements Case

$$U(x_1,x_2) = \min\{ax_1,x_2\}$$



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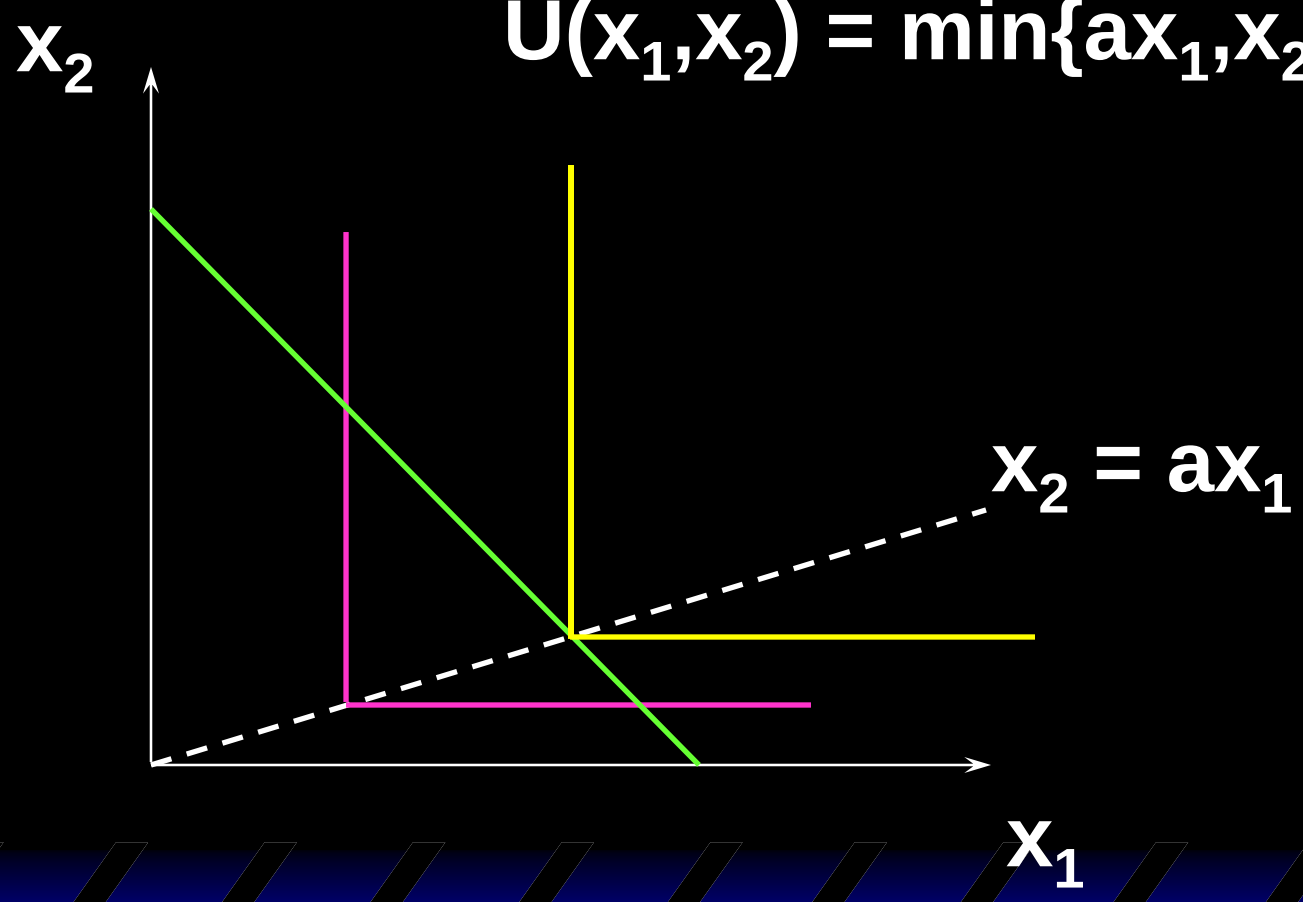

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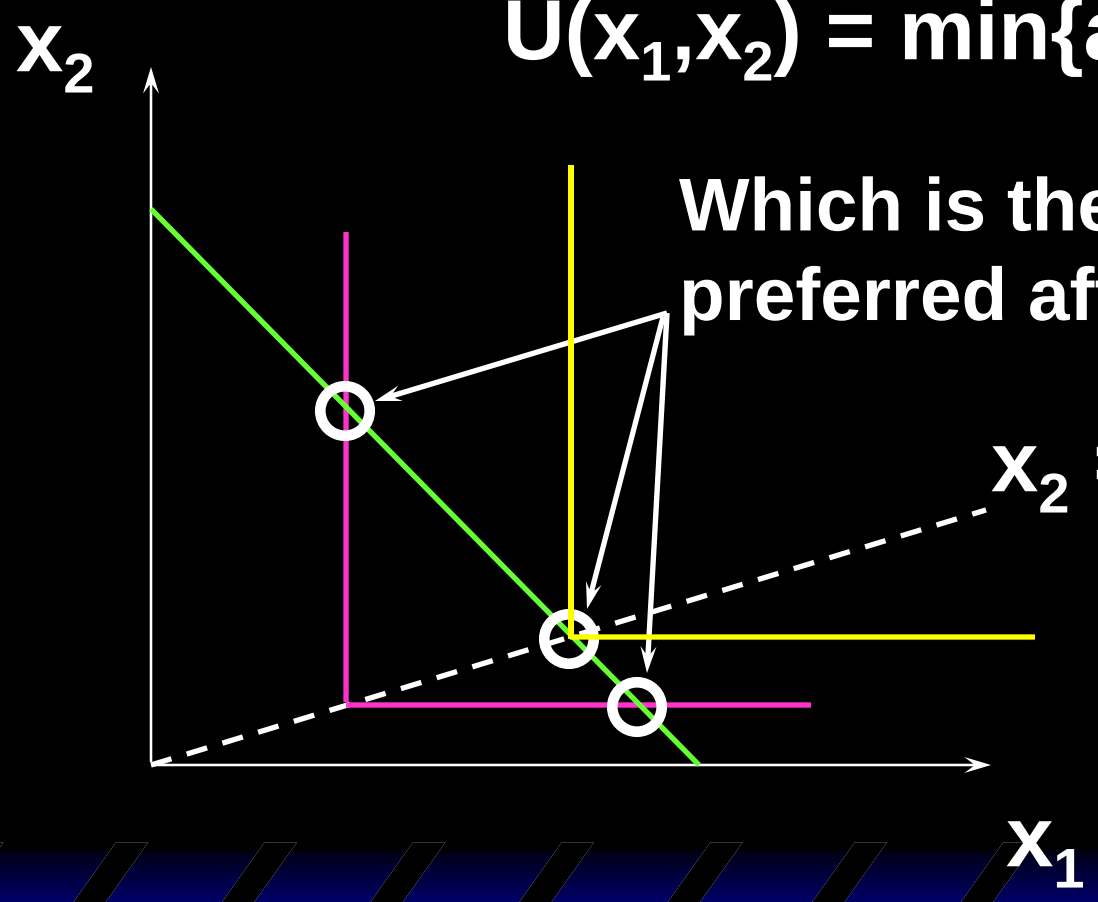


**Which is the most preferred affordable bundle?**

$x_2 = ax_1$

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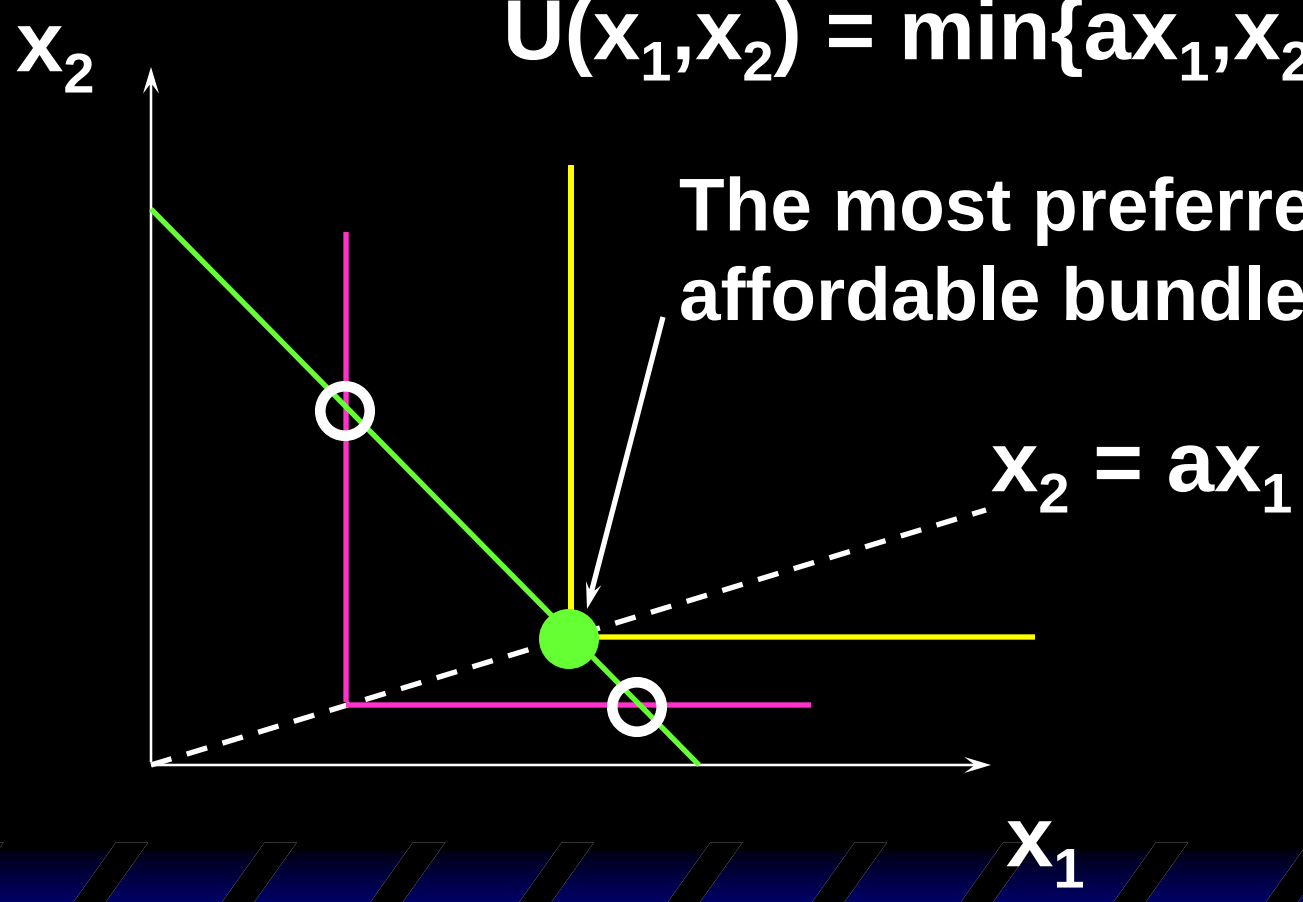

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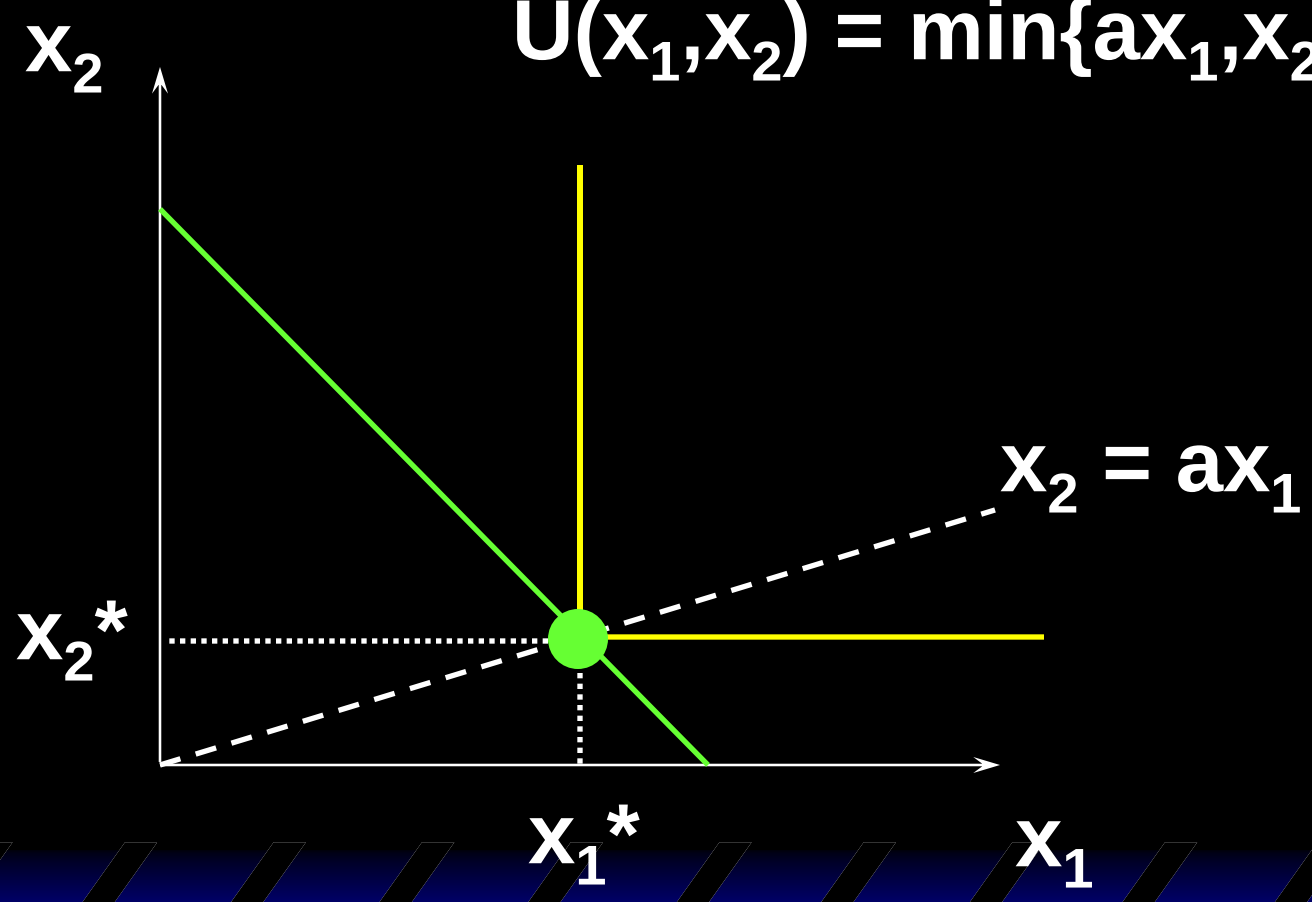

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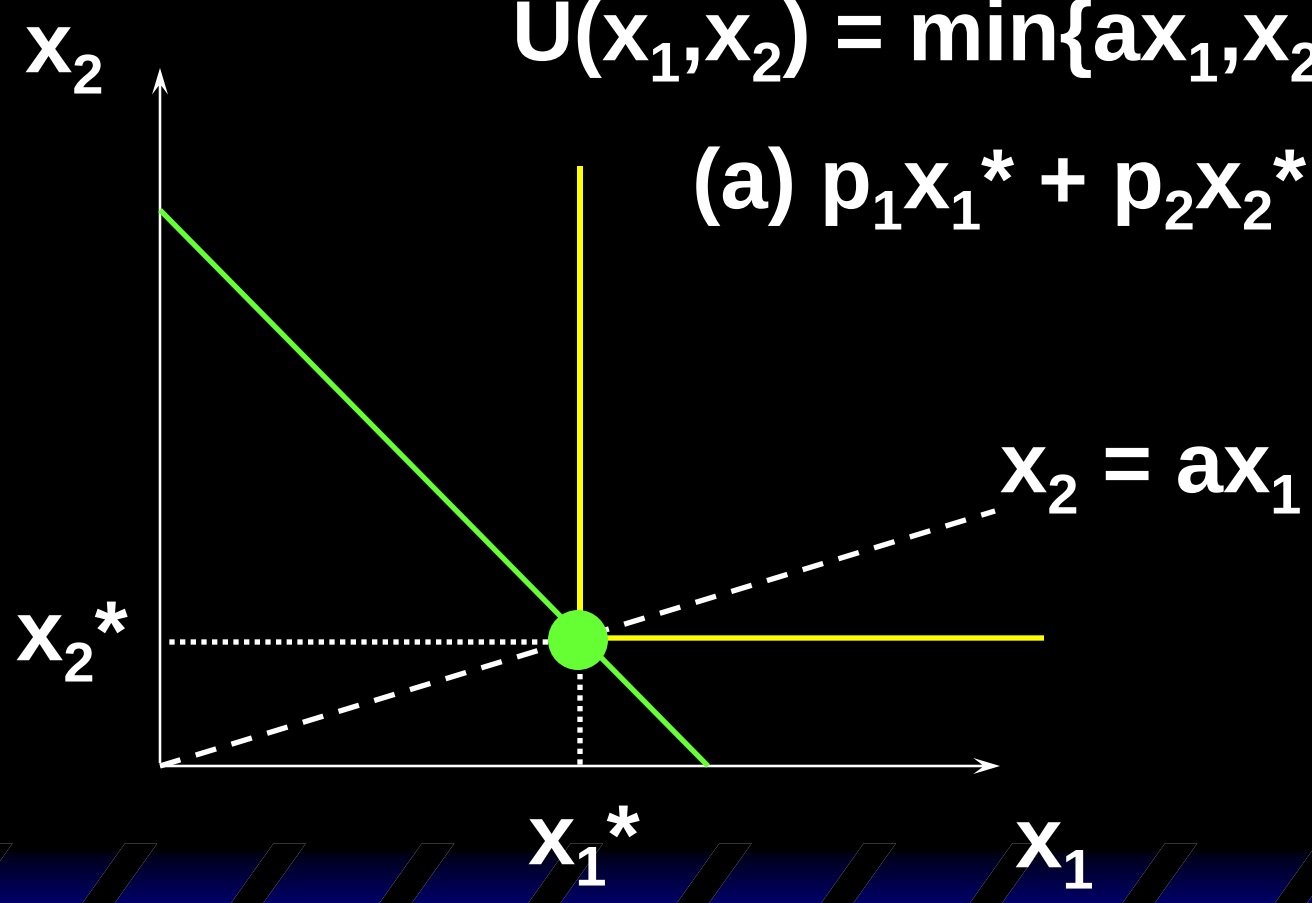
$$\text{(a) } p_1x_1^* + p_2x_2^* = m$$



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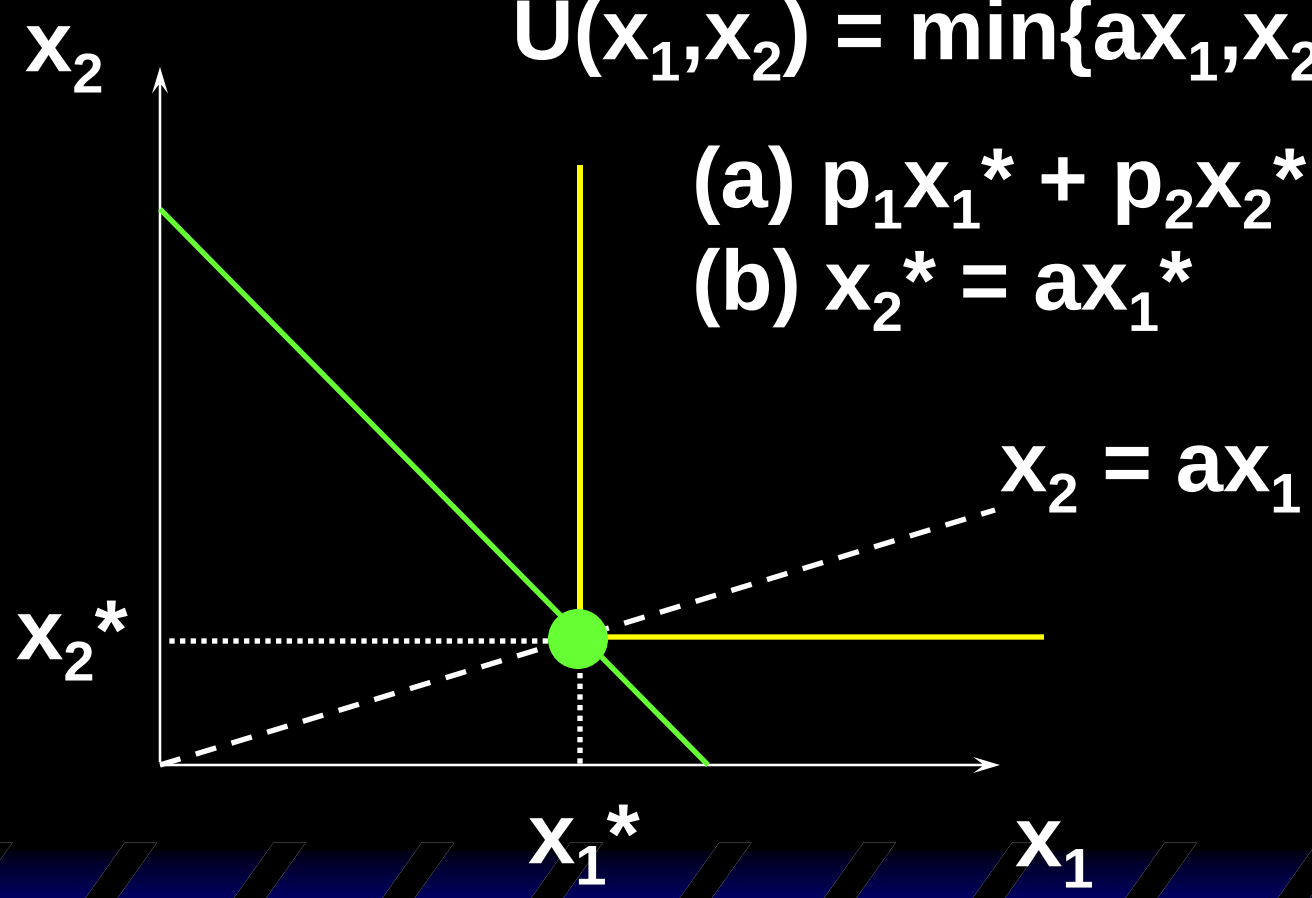
(b) $x_2^* = ax_1^*$



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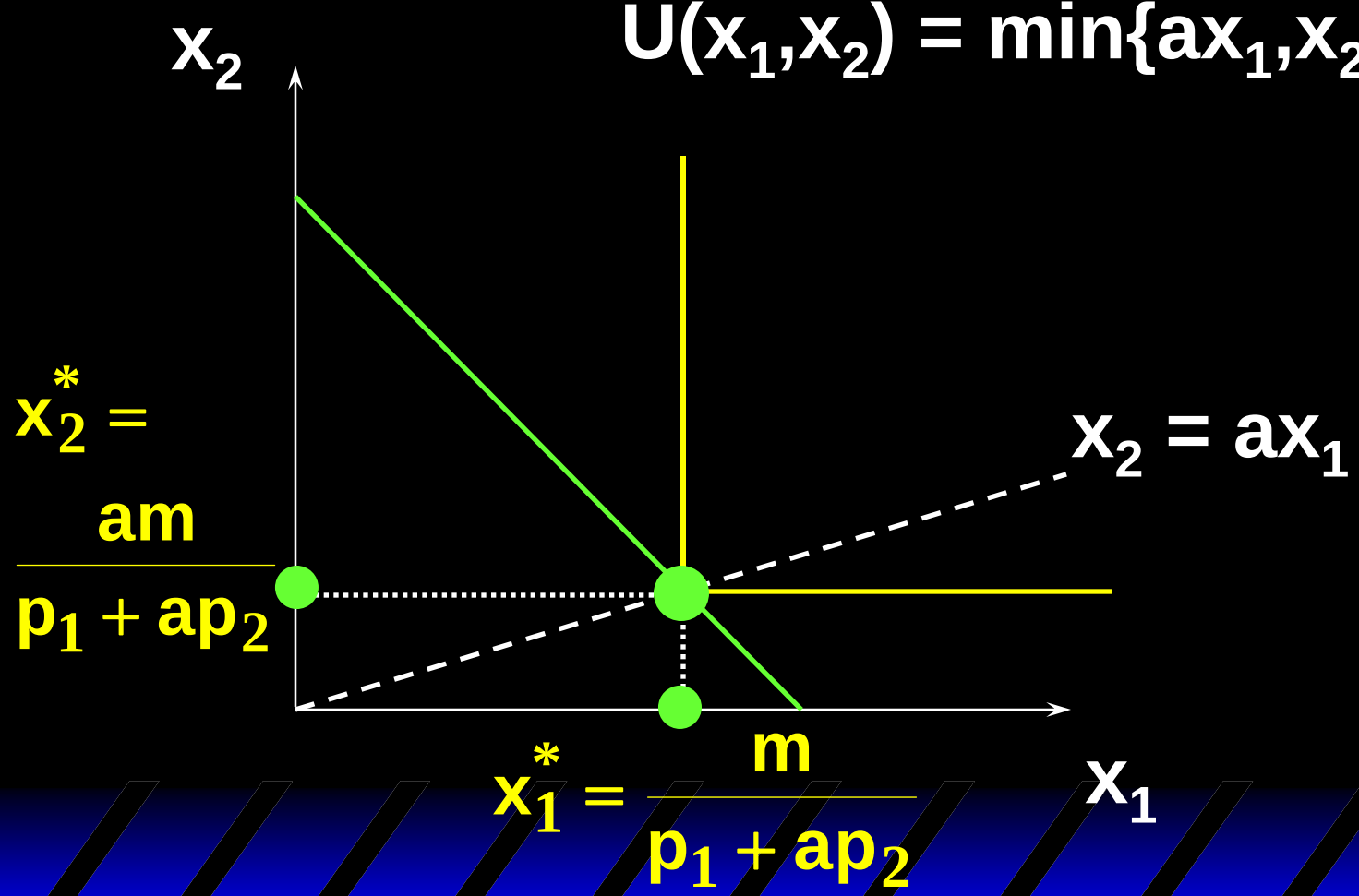
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